MA_vax: vaccination-impact analysis report

Massachusetts, 2025–2026 influenza season (2025-09-01 to 2026-07-04, 307 days). Age-structured SEIR model with a parallel vaccinated arm, calibrated to daily hospital admissions by age group with Bayesian MCMC. This report documents the model, the fit and the vaccination-impact analysis, including scenarios that raise vaccination coverage to 50–70% or cut the season's doses by 5–20%.

Every count is a **hospital admission over the whole 307-day season** unless stated otherwise. Intervals are 95% intervals across 726 simulations, one per posterior parameter draw.

Summary

- The fitted baseline produces **8,268** [7,715 – 8,802] hospitalizations, against **22,545** with no vaccination: the 38.6% overall coverage achieved averts **14,346 [11,362 – 16,901]** hospitalizations (**63.4% [58.0% – 66.7%]**).
- Raising every age group to a coverage target (groups already above it unchanged) averts 2,860 [2,599 – 3,087] more at 50%, 3,589 [3,283 – 3,865] more at 55%, 4,219 [3,874 – 4,532] more at 60%, 4,787 [4,416 – 5,134] more at 65%, 5,279 [4,881 – 5,665] more at 70% (vs. baseline).
- Removing doses from every age group adds 476 [419 – 519] at –5%, 976 [856 – 1,069] at –10%, 1,501 [1,308 – 1,647] at –15%, 2,049 [1,773 – 2,260] at –20% hospitalizations.
- Per dose, vaccinating school-age groups prevents the most hospitalizations across the population (Table S.A.2: 13-17 1,926 and 5-12 1,821 per 100,000 doses), mostly through indirect protection of 65+.

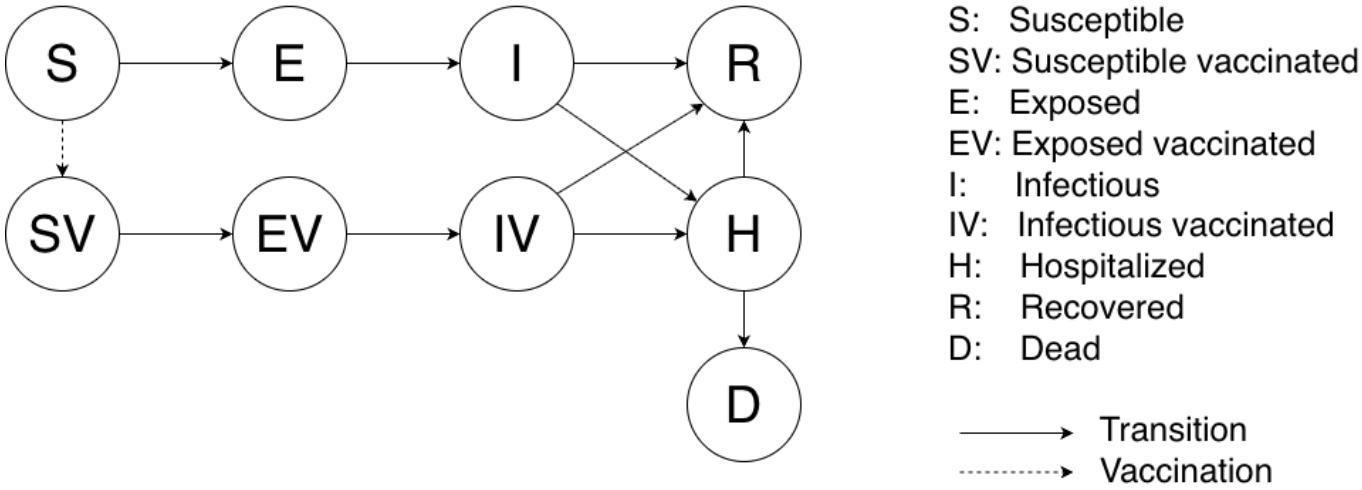
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1. Model structure

1.1 Compartments

Nine compartments, each stratified by 7 age groups (`0`, `1–4`, `5–12`, `13–17`, `18–49`, `50–64`, `65+`):

Compartment	Meaning
`S`	Susceptible, unvaccinated
`E`	Exposed (latent), unvaccinated
`I`	Infectious, unvaccinated
`R`	Recovered (either arm)
`SV`	Susceptible, vaccinated
`EV`	Exposed (latent), vaccinated
`IV`	Infectious, vaccinated
`H`	Hospitalized (either arm)
`D`	Dead (either arm)



Model structure

The vaccinated arm (`SV → EV → IV → H/R`) mirrors the unvaccinated arm (`S → E → I → H/R`) with its own susceptibility and severity parameters. `H`, `R` and `D` are shared, so vaccination history is not tracked once hospitalized or recovered.

1.2 Force of infection

Contacts come from fixed age × age contact matrices (Mistry et al. 2021), with school and work contacts removed on non-school / non-work days:

```
C(t) = total_C − (1 − is_school(t))·school_C − (1 − is_work(t))·work_C
beta_adj(t) = beta_baseline · m(t) · (1 + humidity_impact · exp(−180 · humidity(t)))
wtd_inf_prop(t) = (I·I_relative_infectiousness + IV·IV_relative_infectiousness) / population
foi(t) = beta_adj(t) · (C(t) @ wtd_inf_prop(t))
S_to_E   = foi(t) · relative_suscept   · S
SV_to_EV = foi(t) · vax_susceptibility · SV
```

`m(t)` is a smoothly time-varying transmission multiplier (§2.3) that absorbs behavioural and seasonal variation the mechanistic terms do not capture. `1 − vax_susceptibility` is the vaccine effectiveness (VE) against infection.

1.3 Progression, hospitalization, death

```
E_to_I  = E_to_I_rate · E                EV_to_IV = EV_to_IV_rate · EV
I_to_H  = I_out_rate · I_to_H_prop · I    IV_to_H  = I_out_rate · IV_to_H_prop · IV
I_to_R  = I_out_rate · (1−I_to_H_prop)·I  IV_to_R  = I_out_rate · (1−IV_to_H_prop)·IV
H_to_D  = H_out_rate · H_to_D_prop · H
H_to_R  = H_out_rate · (1−H_to_D_prop)·H
```

1.4 Vaccination flow

Daily doses (an age-specific share of the population, applied 14 days after the dose date to represent the delay to immunity) move an exact count from `S` to `SV` each day, capped at what remains in `S`:

```
S_to_SV(t) = min(round(vax_prop(t − delay) · (S + SV)), S)
```

A dose scheduled for someone who has already been infected is not delivered by the model (the cap). Those doses are still bought and given in reality, so every "per 100,000 doses" figure divides by **scheduled** doses (appendix B).

1.5 Simulation and uncertainty

Explicit Euler integration with 7 sub-steps per day. Every interval comes from a **posterior parameter ensemble**: one deterministic run per posterior draw (726 draws), the same draws for every scenario, so differences between scenarios are paired. Intervals therefore reflect calibration uncertainty, not epidemic-process noise.

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2. Parameters

2.1 Fixed scalar parameters

Parameter	Value	Meaning
`relative_suscept`	1.0	Susceptibility multiplier, unvaccinated arm
`I_relative_infectiousness`	1.0	Infectiousness weight, unvaccinated `I`
`IV_relative_infectiousness`	1.0	Infectiousness weight, vaccinated `IV`
`E_to_I_rate`	0.5	Latent → infectious rate (/day)
`EV_to_IV_rate`	0.5	Same, vaccinated arm (/day)
`I_out_rate`	0.333	Infectious-period exit rate (/day)
`H_out_rate`	0.17	Hospital exit rate (/day)
`vax_transfer_delay_days`	14.0	Days from dose to modelled immunity

2.2 Age-stratified parameters

Age group	Population	Hosp. risk given infection[1]	Hosp. risk given breakthrough[1]	Death risk given hosp.	Residual susceptibility if vaccinated	Initial infections seeded
0	68,723	3.003% (2.211–3.464%) [0.697%]	2.739% (2.016–3.159%) [0.636%]	1.74%	0.57	2
1-4	287,051	2.275% (1.332–3.308%) [0.697%]	2.075% (1.215–3.017%) [0.636%]	1.74%	0.57	8
5-12	625,439	0.781% (0.459–1.220%) [0.274%]	0.712% (0.418–1.113%) [0.250%]	1.17%	0.57	18
13-17	406,983	0.387% (0.210–0.596%) [0.274%]	0.353% (0.191–0.543%) [0.250%]	1.17%	0.57	12
18-49	3,112,693	0.462% (0.277–0.716%) [0.561%]	0.421% (0.252–0.652%) [0.511%]	2.63%	0.79	89
50-64	1,351,323	1.443% (0.866–2.202%) [1.060%]	1.314% (0.789–2.006%) [0.966%]	6.30%	0.79	39
65+	1,420,722	11.966% (7.229–17.253%) [9.091%]	8.257% (4.988–11.904%) [6.273%]	7.99%	1.00	41

[1] Fitted: pre-fit value × the age group's fitted multiplier (§2.4), posterior mean (95% interval), pre-fit value in brackets. Both columns share the multiplier, so VE against hospitalization given infection is unchanged by the fit. Seeds are pre-fit values, scaled by one fitted multiplier. A residual susceptibility of 1.00 (65+) means vaccination blocks no infection there: it only reduces severity, and vaccinating 65+ has no indirect effect on other age groups.

2.3 Fitting method

Affine-invariant ensemble MCMC (64 walkers, 4000 iterations each) over the 307-day target window.

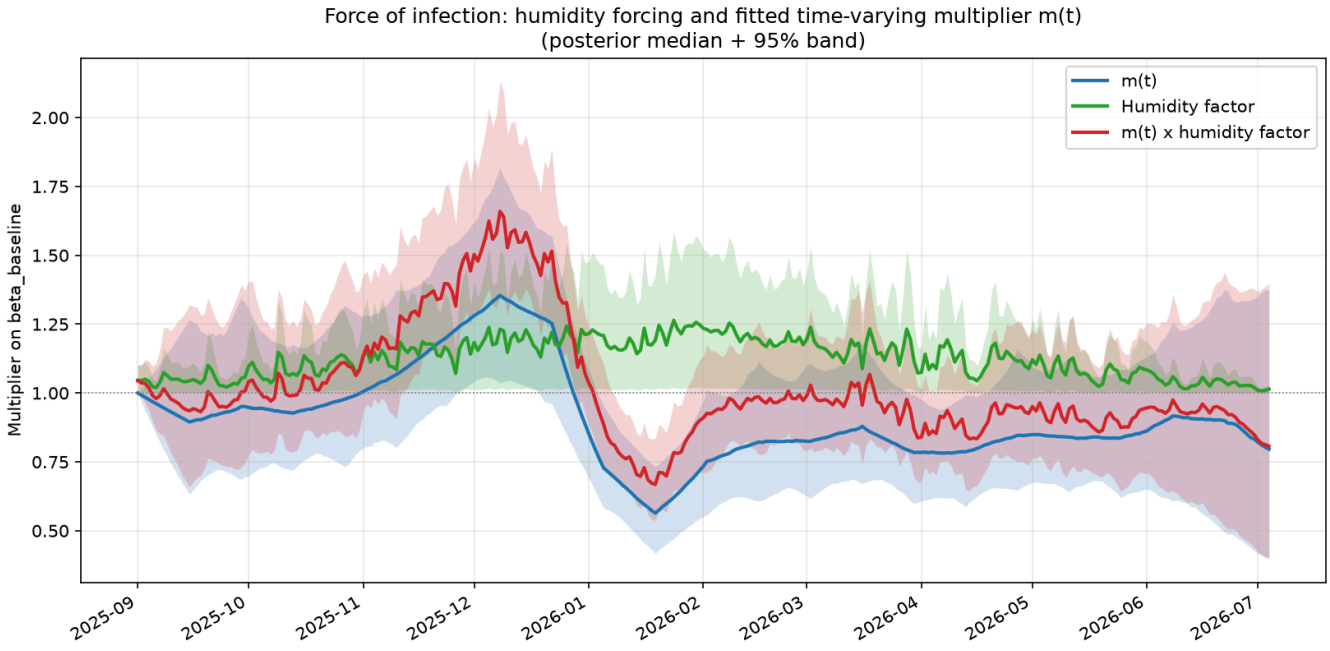
Parameter	Prior	Role
`beta_baseline`	Uniform(0.015, 0.06)	Baseline transmission rate
`humidity_impact`	Uniform(0, 1)	Strength of humidity forcing
Initial-seed multiplier	Log-uniform(0.1, 10.0)	Multiplies the age-specific seed counts
Hospitalization-risk multiplier (× 7)	Uniform(0.1, 5.0)	Multiplies both hospitalization-risk columns of one age group
`m(t)` log-increments (× 22)	Normal(0, 0.25)	Random-walk steps in log-transmission, knots every 14 days
`phi`	—	Negative-binomial dispersion (likelihood only)

Likelihood: negative-binomial (NB2), jointly over 8 targets: daily admissions for each age group (weights 0_0: 1, 1_4: 1, 5_12: 1, 13_17: 1, 18_49: 2, 50_64: 2, 65plus: 5) and end-of-season cumulative admissions by age (weight 10).

Posterior sample: first 1700 iterations discarded, then every 200th kept, giving **726 draws**. "Best" is the draw with the highest log-posterior.

2.4 Fitted parameters

Parameter	Posterior mean	5%	95%	"Best"
beta_baseline	0.0317	0.0273	0.0357	0.0330
humidity_impact	0.31	0.0512	0.61	0.46
Initial-seed multiplier	1.76	0.63	3.45	2.12
Hospitalization-risk multiplier – 0	4.31	3.35	4.94	4.46
Hospitalization-risk multiplier – 1-4	3.26	2.08	4.61	2.75
Hospitalization-risk multiplier – 5-12	2.85	1.87	4.19	2.26
Hospitalization-risk multiplier – 13-17	1.41	0.85	2.03	1.30
Hospitalization-risk multiplier – 18-49	0.82	0.53	1.15	0.55
Hospitalization-risk multiplier – 50-64	1.36	0.88	1.94	1.08
Hospitalization-risk multiplier – 65+	1.32	0.90	1.79	1.08
phi (NB dispersion)	112	29	324	106

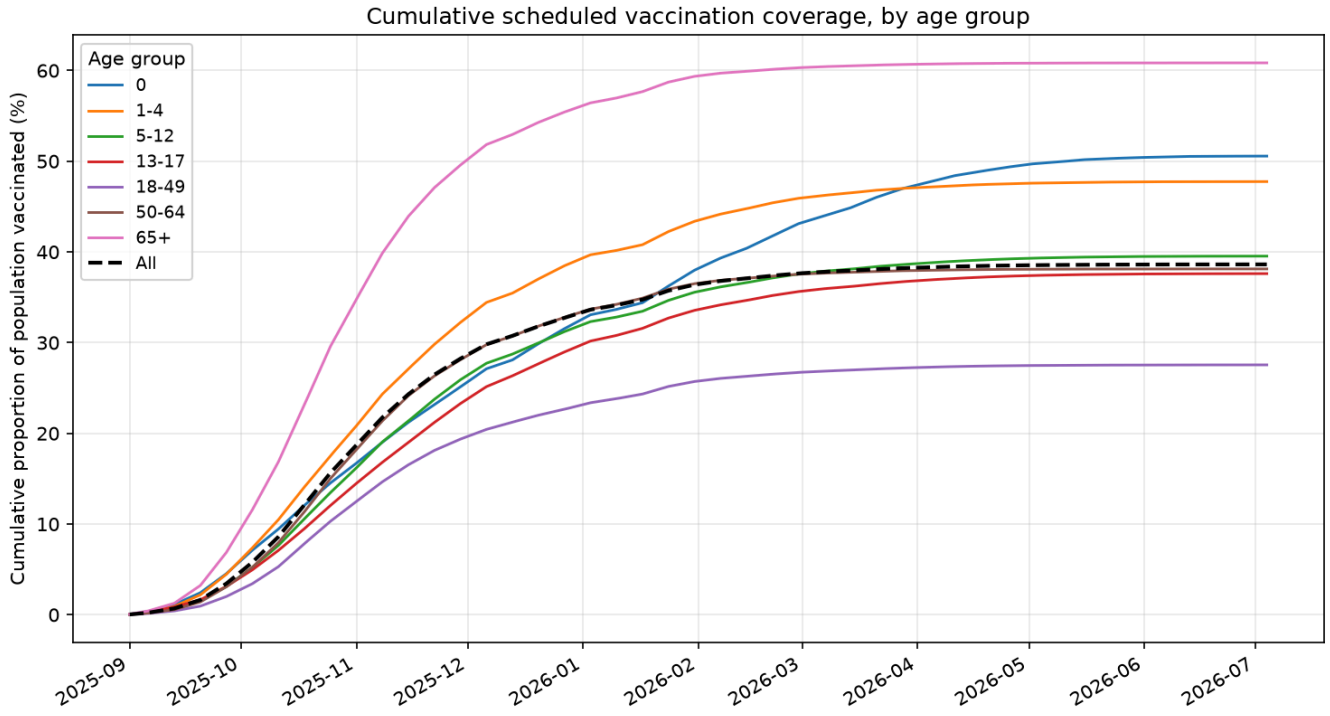


Humidity factor, fitted m(t) and their product (posterior median and 95% band)

2.5 Vaccination coverage

Cumulative share of each age group vaccinated by the schedule over the simulated season (after the 14-day delay):

Age group	Population	Scheduled coverage
0	68,723	50.6%
1-4	287,051	47.8%
5-12	625,439	39.5%
13-17	406,983	37.6%
18-49	3,112,693	27.6%
50-64	1,351,323	38.1%
65+	1,420,722	60.8%
All	**7,272,934**	**38.6%**

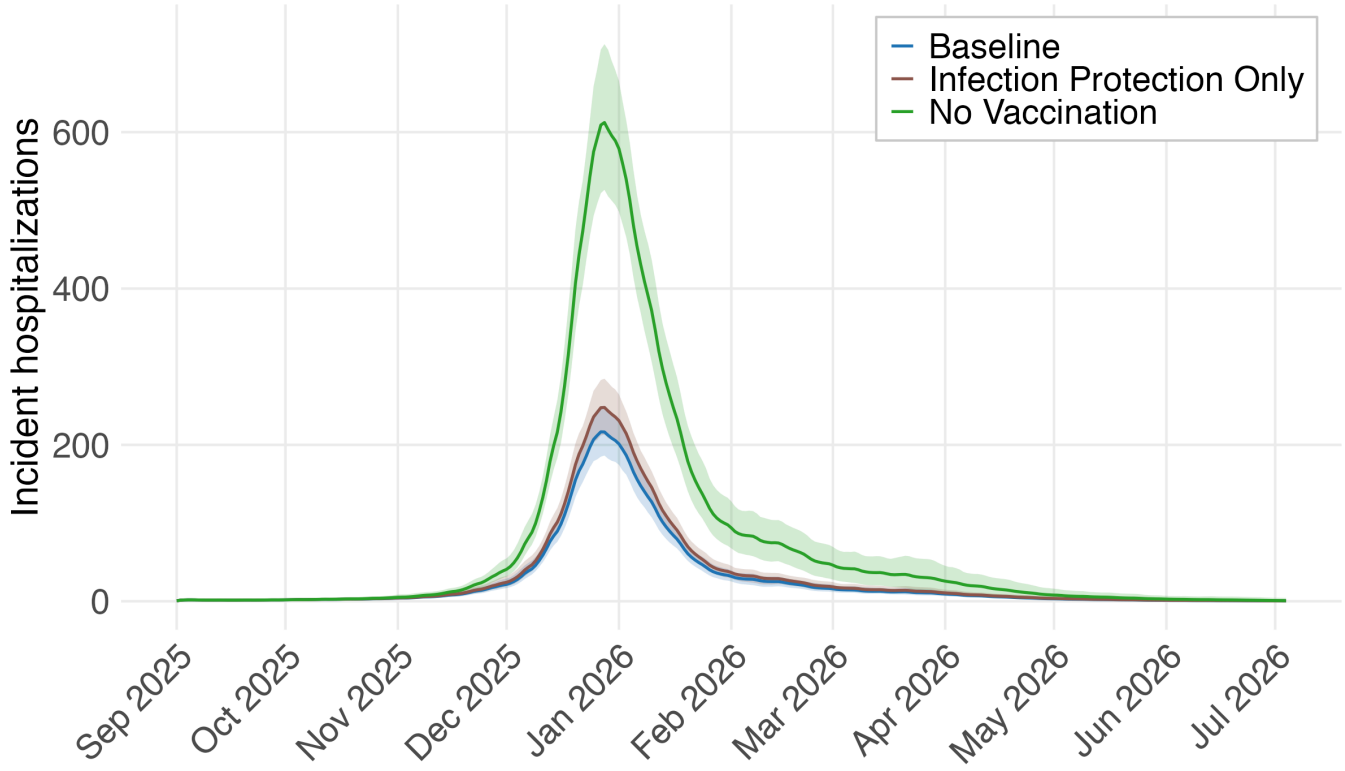


Cumulative scheduled vaccination coverage

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3. Vaccination impact of the 2025–2026 programme

The fitted baseline produces a median of **8,268** hospitalizations across all ages, against **22,545** with no vaccination.



Daily hospitalizations: baseline, infection protection only, no vaccination

Table S.A.1 — Hospitalizations averted, infection vs. severity protection

Infection protection (no vaccination → VE against infection only)

Age group	Hospitalizations averted	% averted	Per 100,000 population	Per 100,000 doses
0	150 [108 – 187]	63.4% [58.0% – 66.7%]	218.0 [156.8 – 272.5]	431.0 [310.0 – 538.9]
1-4	578 [397 – 746]	64.5% [58.9% – 68.0%]	201.2 [138.2 – 259.8]	421.3 [289.5 – 543.9]
5-12	598 [411 – 851]	62.8% [56.3% – 66.8%]	95.7 [65.7 – 136.1]	242.0 [166.2 – 344.1]
13-17	199 [126 – 274]	61.8% [55.1% – 65.9%]	48.9 [31.0 – 67.4]	130.0 [82.5 – 179.2]
18-49	1,396 [1,023 – 1,859]	58.4% [51.8% – 62.3%]	44.9 [32.9 – 59.7]	162.8 [119.2 – 216.7]
50-64	1,672 [1,225 – 2,199]	59.5% [53.3% – 63.2%]	123.7 [90.6 – 162.7]	324.5 [237.7 – 426.8]
65+	8,492 [6,508 – 10,274]	57.2% [51.1% – 60.9%]	597.7 [458.1 – 723.2]	982.4 [752.9 – 1188.6]
All	**13,178 [10,271 – 15,701]**	**58.2% [52.1% – 62.0%]**	**181.2 [141.2 – 215.9]**	**469.1 [365.6 – 558.9]**

Severity protection (adding VE against hospitalization given infection)

Age group	Hospitalizations averted	% averted	Per 100,000 population	Per 100,000 doses
0	2 [1 – 2]	0.7% [0.6% – 0.8%]	2.4 [1.5 – 3.5]	4.7 [3.1 – 6.9]
1-4	7 [5 – 9]	0.8% [0.7% – 1.0%]	2.6 [1.8 – 3.2]	5.4 [3.8 – 6.7]
5-12	7 [5 – 8]	0.7% [0.6% – 0.8%]	1.1 [0.8 – 1.3]	2.7 [1.9 – 3.4]
13-17	2 [1 – 3]	0.6% [0.6% – 0.8%]	0.5 [0.3 – 0.7]	1.3 [0.9 – 1.7]
18-49	17 [14 – 21]	0.7% [0.6% – 0.8%]	0.5 [0.4 – 0.7]	2.0 [1.6 – 2.4]
50-64	29 [24 – 35]	1.0% [0.9% – 1.2%]	2.1 [1.8 – 2.6]	5.5 [4.6 – 6.7]
65+	1,098 [991 – 1,193]	7.4% [6.7% – 8.4%]	77.3 [69.8 – 84.0]	127.0 [114.7 – 138.0]
All	**1,161 [1,056 – 1,255]**	**5.1% [4.6% – 5.9%]**	**16.0 [14.5 – 17.3]**	**41.3 [37.6 – 44.7]**

Total (no vaccination → baseline)

Age group	Hospitalizations averted	% averted	Per 100,000 population	Per 100,000 doses
0	151 [109 – 189]	64.0% [58.8% – 67.3%]	220.3 [158.3 – 275.7]	435.8 [313.1 – 545.3]
1-4	585 [402 – 754]	65.3% [59.9% – 68.7%]	203.8 [140.2 – 262.8]	426.9 [293.6 – 550.3]
5-12	605 [417 – 859]	63.5% [57.1% – 67.4%]	96.8 [66.7 – 137.4]	244.7 [168.6 – 347.5]
13-17	201 [128 – 277]	62.5% [55.9% – 66.5%]	49.4 [31.4 – 68.1]	131.3 [83.5 – 181.0]
18-49	1,414 [1,039 – 1,879]	59.1% [52.6% – 63.0%]	45.4 [33.4 – 60.4]	164.9 [121.2 – 219.1]
50-64	1,700 [1,250 – 2,233]	60.5% [54.4% – 64.1%]	125.8 [92.5 – 165.3]	330.0 [242.6 – 433.5]
65+	9,585 [7,553 – 11,402]	64.5% [59.5% – 67.6%]	674.6 [531.7 – 802.5]	1108.8 [873.8 – 1319.0]
All	**14,346 [11,362 – 16,901]**	**63.4% [58.0% – 66.7%]**	**197.3 [156.2 – 232.4]**	**510.7 [404.4 – 601.6]**

Table S.A.2 — Hospitalizations averted by age group vaccinated

Each column vaccinates one age group only (all others unvaccinated) and compares with no vaccination; `All` is the full baseline schedule. Rows are the age group in which hospitalizations are counted, so off-diagonal cells are indirect protection.

Hospitalizations averted

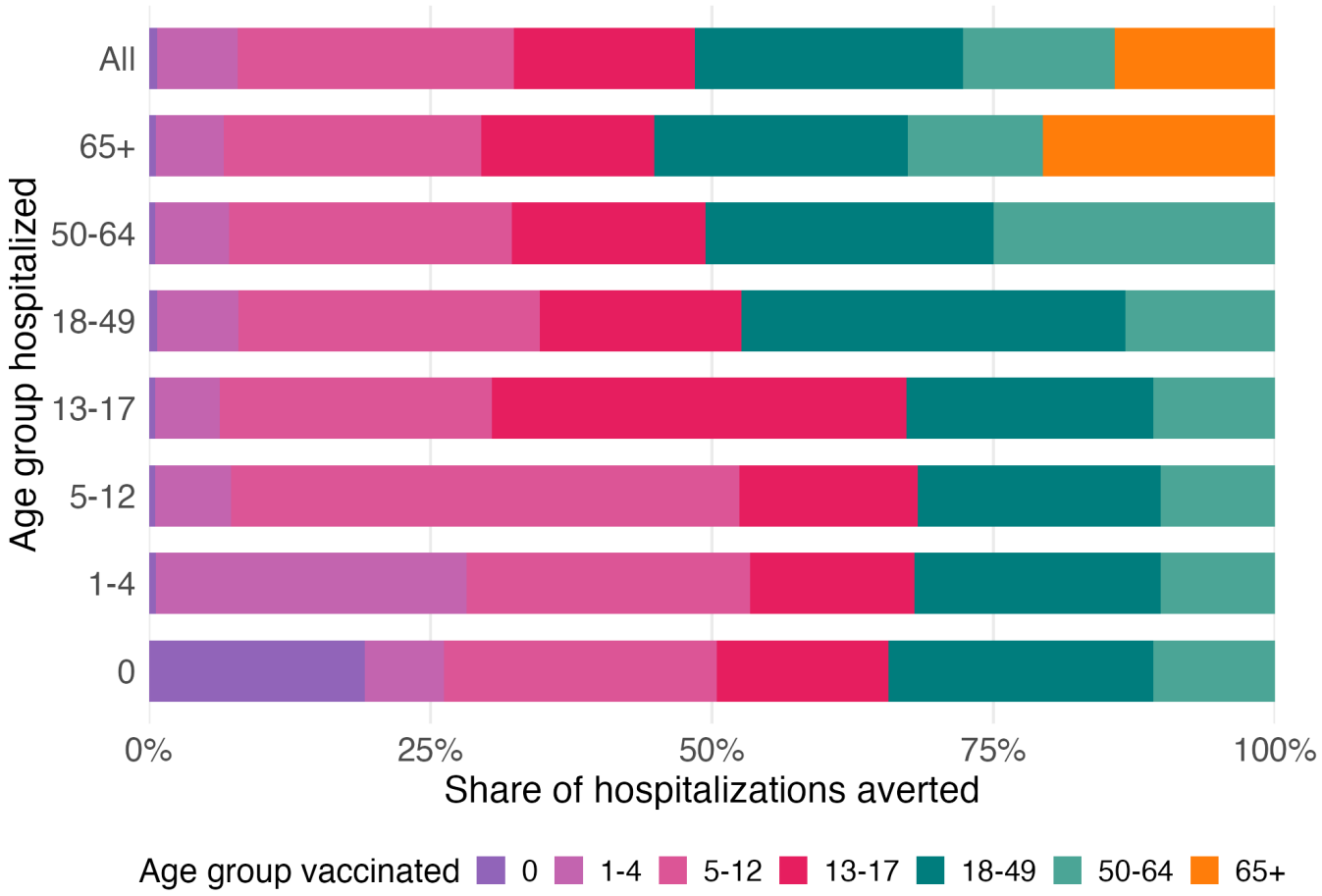
Counted in	0 vacc.	1-4 vacc.	5-12 vacc.	13-17 vacc.	18-49 vacc.	50-64 vacc.	65+ vacc.	All vacc.
0	38 [26 – 48]	14 [10 – 17]	48 [34 – 57]	30 [21 – 36]	46 [33 – 56]	21 [15 – 25]	0 [-0 – 0]	151 [109 – 189]
1-4	5 [3 – 6]	207 [144 – 266]	189 [124 – 253]	109 [70 – 148]	165 [108 – 220]	77 [49 – 103]	0 [-0 – 0]	585 [402 – 754]
5-12	4 [3 – 6]	48 [31 – 72]	325 [223 – 468]	114 [72 – 173]	155 [99 – 232]	73 [46 – 111]	0 [-0 – 0]	605 [417 – 859]
13-17	1 [1 – 2]	14 [8 – 20]	59 [34 – 85]	89 [57 – 123]	53 [31 – 77]	26 [15 – 38]	0 [-0 – 0]	201 [128 – 277]
18-49	12 [8 – 16]	120 [82 – 170]	449 [302 – 631]	299 [199 – 425]	569 [408 – 778]	221 [151 – 310]	0 [-0 – 0]	1,414 [1,039 – 1,879]
50-64	13 [9 – 18]	135 [92 – 189]	524 [354 – 726]	358 [241 – 499]	531 [364 – 733]	515 [379 – 680]	0 [-0 – 0]	1,700 [1,250 – 2,233]
65+	71 [51 – 92]	757 [544 – 971]	2,877 [2,049 – 3,671]	1,927 [1,364 – 2,479]	2,824 [2,045 – 3,577]	1,507 [1,101 – 1,907]	2,586 [2,184 – 2,942]	9,585 [7,553 – 11,402]
All	**144 [117 – 167]**	**1,297 [954 – 1,632]**	**4,504 [3,241 – 5,672]**	**2,948 [2,094 – 3,736]**	**4,371 [3,217 – 5,483]**	**2,465 [1,846 – 3,065]**	**2,586 [2,184 – 2,942]**	**14,346 [11,362 – 16,901]**

% reduction

Counted in	0 vacc.	1-4 vacc.	5-12 vacc.	13-17 vacc.	18-49 vacc.	50-64 vacc.	65+ vacc.	All vacc.
0	15.8% [15.2% – 16.3%]	5.8% [5.0% – 6.6%]	20.0% [16.6% – 22.5%]	12.6% [10.3% – 14.4%]	19.4% [16.3% – 21.7%]	8.9% [7.4% – 10.1%]	0.0% [-0.0% – 0.0%]	64.0% [58.8% – 67.3%]
1-4	0.5% [0.4% – 0.6%]	23.1% [21.9% – 24.1%]	21.1% [17.4% – 23.9%]	12.2% [9.8% – 14.1%]	18.3% [15.1% – 20.8%]	8.5% [7.0% – 9.8%]	0.0% [-0.0% – 0.0%]	65.3% [59.9% – 68.7%]
5-12	0.4% [0.3% – 0.5%]	5.1% [4.1% – 5.9%]	34.2% [30.3% – 37.1%]	12.0% [9.4% – 14.2%]	16.3% [12.9% – 19.0%]	7.7% [6.1% – 9.1%]	0.0% [-0.0% – 0.0%]	63.5% [57.1% – 67.4%]
13-17	0.4% [0.3% – 0.5%]	4.3% [3.3% – 5.1%]	18.1% [14.3% – 21.2%]	27.5% [24.6% – 29.9%]	16.4% [13.0% – 19.1%]	8.1% [6.4% – 9.4%]	0.0% [-0.0% – 0.0%]	62.5% [55.9% – 66.5%]
18-49	0.5% [0.4% – 0.6%]	5.0% [4.1% – 5.7%]	18.6% [15.1% – 21.4%]	12.4% [9.9% – 14.3%]	23.7% [20.5% – 26.2%]	9.2% [7.5% – 10.5%]	0.0% [-0.0% – 0.0%]	59.1% [52.6% – 63.0%]
50-64	0.4% [0.4% – 0.5%]	4.8% [3.9% – 5.5%]	18.4% [15.0% – 21.1%]	12.6% [10.2% – 14.5%]	18.7% [15.5% – 21.1%]	18.3% [16.6% – 19.6%]	0.0% [-0.0% – 0.0%]	60.5% [54.4% – 64.1%]
65+	0.5% [0.4% – 0.5%]	5.1% [4.3% – 5.8%]	19.3% [16.1% – 21.8%]	13.0% [10.7% – 14.7%]	19.0% [16.0% – 21.3%]	10.1% [8.6% – 11.3%]	17.4% [17.2% – 17.5%]	64.5% [59.5% – 67.6%]
All	**0.6% [0.6% – 0.7%]**	**5.8% [4.9% – 6.5%]**	**19.8% [16.5% – 22.4%]**	**13.0% [10.7% – 14.8%]**	**19.3% [16.2% – 21.7%]**	**10.9% [9.3% – 12.1%]**	**11.5% [10.8% – 12.0%]**	**63.4% [58.0% – 66.7%]**

Per 100,000 doses given to the column's age group

Counted in	0 vacc.	1-4 vacc.	5-12 vacc.	13-17 vacc.	18-49 vacc.	50-64 vacc.	65+ vacc.	All vacc.
0	107.9 [76.2 – 137.4]	10.1 [7.2 – 12.2]	19.2 [13.7 – 23.0]	19.6 [14.0 – 23.2]	5.4 [3.8 – 6.5]	4.1 [2.9 – 4.9]	0.0 [-0.0 – 0.0]	435.8 [313.1 – 545.3]
1-4	13.8 [9.1 – 18.5]	150.9 [104.9 – 194.0]	76.5 [50.2 – 102.2]	71.4 [45.7 – 96.6]	19.2 [12.5 – 25.7]	14.9 [9.6 – 20.0]	0.0 [-0.0 – 0.0]	426.9 [293.6 – 550.3]
5-12	11.6 [7.4 – 17.5]	35.1 [22.6 – 52.5]	131.6 [90.2 – 189.4]	74.6 [46.8 – 113.1]	18.1 [11.5 – 27.1]	14.2 [9.0 – 21.5]	0.0 [-0.0 – 0.0]	244.7 [168.6 – 347.5]
13-17	3.7 [2.1 – 5.4]	10.1 [5.8 – 14.9]	23.7 [13.7 – 34.5]	58.0 [37.1 – 80.3]	6.2 [3.6 – 9.0]	5.1 [3.0 – 7.4]	0.0 [-0.0 – 0.0]	131.3 [83.5 – 181.0]
18-49	33.4 [22.7 – 47.0]	87.9 [59.5 – 123.8]	181.6 [122.1 – 255.4]	195.1 [130.0 – 277.5]	66.3 [47.6 – 90.7]	42.9 [29.2 – 60.2]	0.0 [-0.0 – 0.0]	164.9 [121.2 – 219.1]
50-64	36.3 [24.5 – 50.9]	98.7 [66.9 – 138.0]	212.1 [143.0 – 293.7]	233.6 [157.2 – 326.3]	61.9 [42.5 – 85.5]	99.9 [73.6 – 132.0]	0.0 [-0.0 – 0.0]	330.0 [242.6 – 433.5]
65+	204.4 [147.6 – 263.4]	552.2 [396.6 – 708.0]	1163.6 [828.7 – 1484.7]	1259.2 [891.2 – 1619.4]	329.3 [238.5 – 417.1]	292.5 [213.7 – 370.0]	299.2 [252.6 – 340.3]	1108.8 [873.8 – 1319.0]
All	**414.4 [335.7 – 481.0]**	**946.2 [696.0 – 1190.8]**	**1821.3 [1310.7 – 2293.8]**	**1926.1 [1368.3 – 2440.6]**	**509.7 [375.1 – 639.3]**	**478.5 [358.4 – 594.8]**	**299.2 [252.6 – 340.3]**	**510.7 [404.4 – 601.6]**



Share of each age group's averted hospitalizations, by age group vaccinated

Table S.A.4 — Vaccine-effectiveness sensitivity scenarios

Scenario	Age group	VE_infection	VE_hosp_infection	VE_hosp_given_infection	VE_transmission_blocking
low_ve	0	33%	33%	0%	0%
low_ve	1-4	33%	33%	0%	0%
low_ve	5-12	33%	33%	0%	0%
low_ve	13-17	33%	33%	0%	0%
low_ve	18-49	6%	10%	4%	0%
low_ve	50-64	6%	10%	4%	0%
low_ve	65+	0%	21%	21%	0%
baseline_ve	0	43%	48%	9%	0%
baseline_ve	1-4	43%	48%	9%	0%
baseline_ve	5-12	43%	48%	9%	0%
baseline_ve	13-17	43%	48%	9%	0%
baseline_ve	18-49	21%	28%	9%	0%
baseline_ve	50-64	21%	28%	9%	0%
baseline_ve	65+	0%	31%	31%	0%
high_ve	0	51%	78%	55%	0%
high_ve	1-4	51%	78%	55%	0%
high_ve	5-12	51%	78%	55%	0%
high_ve	13-17	51%	78%	55%	0%
high_ve	18-49	34%	44%	16%	0%
high_ve	50-64	34%	44%	16%	0%
high_ve	65+	14%	39%	29%	0%

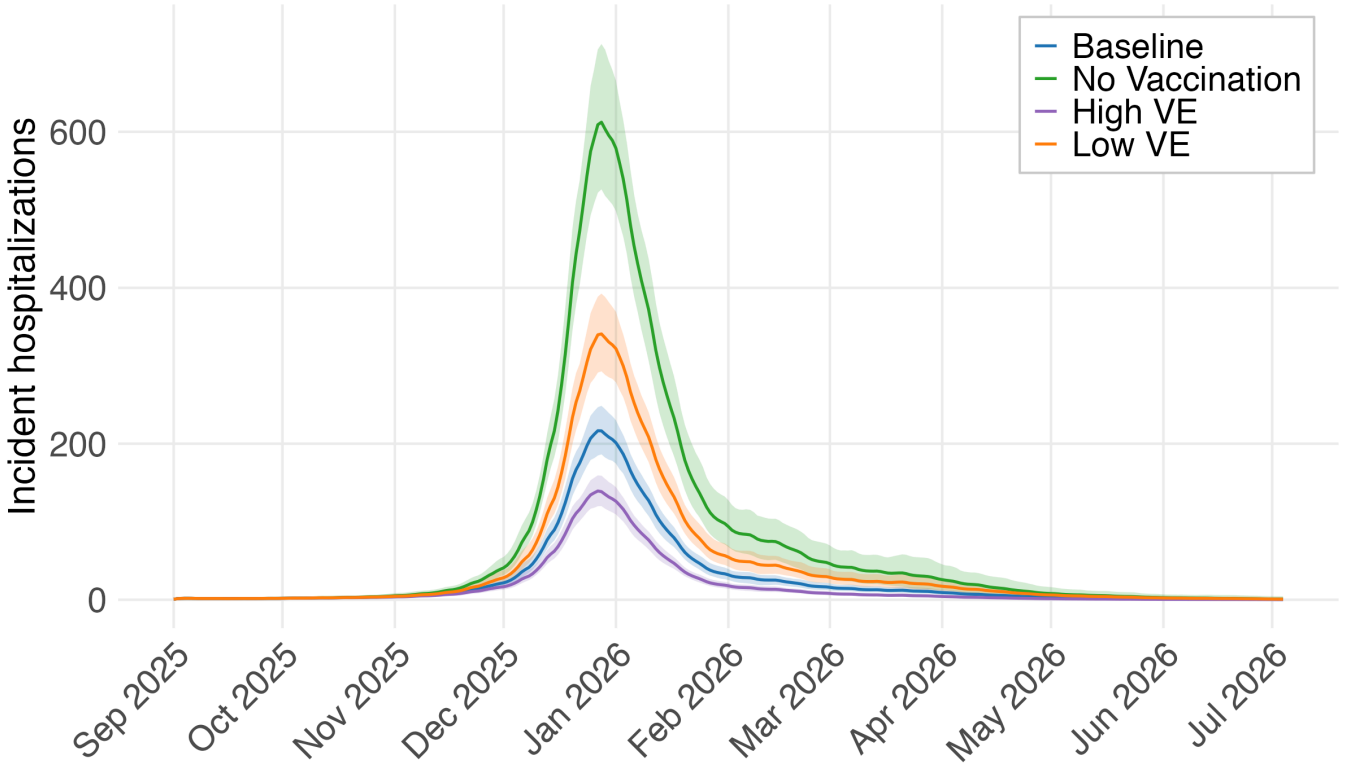
Table S.A.5 — Hospitalizations averted across VE scenarios (vs. no vaccination)

Hospitalizations averted

Age group	Low VE	Baseline VE	High VE
0	101 [71 – 125]	151 [109 – 189]	186 [131 – 235]
1-4	399 [270 – 521]	585 [402 – 754]	714 [499 – 913]
5-12	408 [277 – 591]	605 [417 – 859]	743 [517 – 1,033]
13-17	135 [84 – 188]	201 [128 – 277]	248 [162 – 336]
18-49	864 [605 – 1,190]	1,414 [1,039 – 1,879]	1,764 [1,337 – 2,302]
50-64	1,036 [727 – 1,405]	1,700 [1,250 – 2,233]	2,115 [1,596 – 2,738]
65+	6,409 [4,888 – 7,807]	9,585 [7,553 – 11,402]	11,563 [9,403 – 13,462]
All	**9,420 [7,195 – 11,405]**	**14,346 [11,362 – 16,901]**	**17,393 [14,179 – 20,176]**

% reduction

Age group	Low VE	Baseline VE	High VE
0	42.9% [37.9% – 46.3%]	64.0% [58.8% – 67.3%]	78.5% [74.8% – 80.8%]
1-4	44.4% [39.2% – 48.0%]	65.3% [59.9% – 68.7%]	79.8% [76.1% – 82.1%]
5-12	42.9% [37.0% – 47.1%]	63.5% [57.1% – 67.4%]	77.8% [73.1% – 80.5%]
13-17	41.7% [35.6% – 45.9%]	62.5% [55.9% – 66.5%]	76.9% [72.1% – 79.8%]
18-49	36.1% [30.3% – 40.0%]	59.1% [52.6% – 63.0%]	73.6% [68.5% – 76.6%]
50-64	36.7% [31.2% – 40.5%]	60.5% [54.4% – 64.1%]	75.1% [70.5% – 77.8%]
65+	43.1% [38.5% – 46.4%]	64.5% [59.5% – 67.6%]	77.8% [74.0% – 80.1%]
All	**41.6% [36.6% – 45.0%]**	**63.4% [58.0% – 66.7%]**	**77.1% [73.0% – 79.6%]**



Daily hospitalizations under low, baseline and high VE

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4. Coverage scenarios

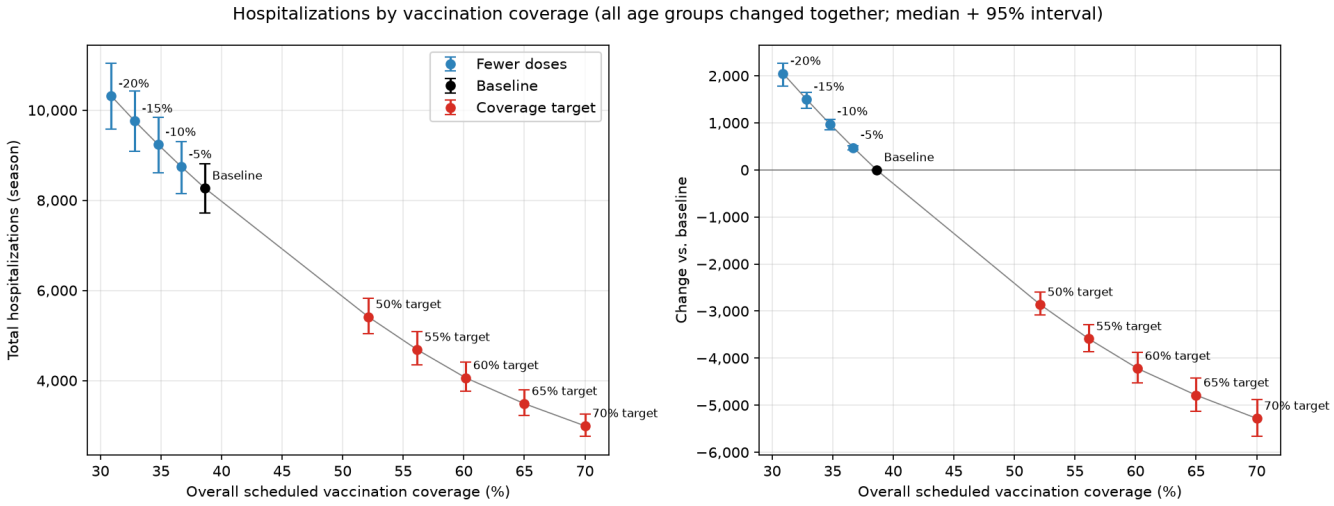
Two families of counterfactual schedules, each applied either to **every age group at once** or to **one age group at a time** (others keep the baseline schedule), and compared with the baseline:

- **Coverage targets** (50%, 55%, 60%, 65%, 70%): each targeted group's daily doses are scaled by `target / baseline coverage`, so its season coverage reaches the target. A group already above the target keeps its schedule.
- **Fewer doses** (–5%, –10%, –15%, –20%): each targeted group's daily doses are cut by that share (a relative decrease, e.g. 60% coverage becomes 54% at –10%). Timing is unchanged.

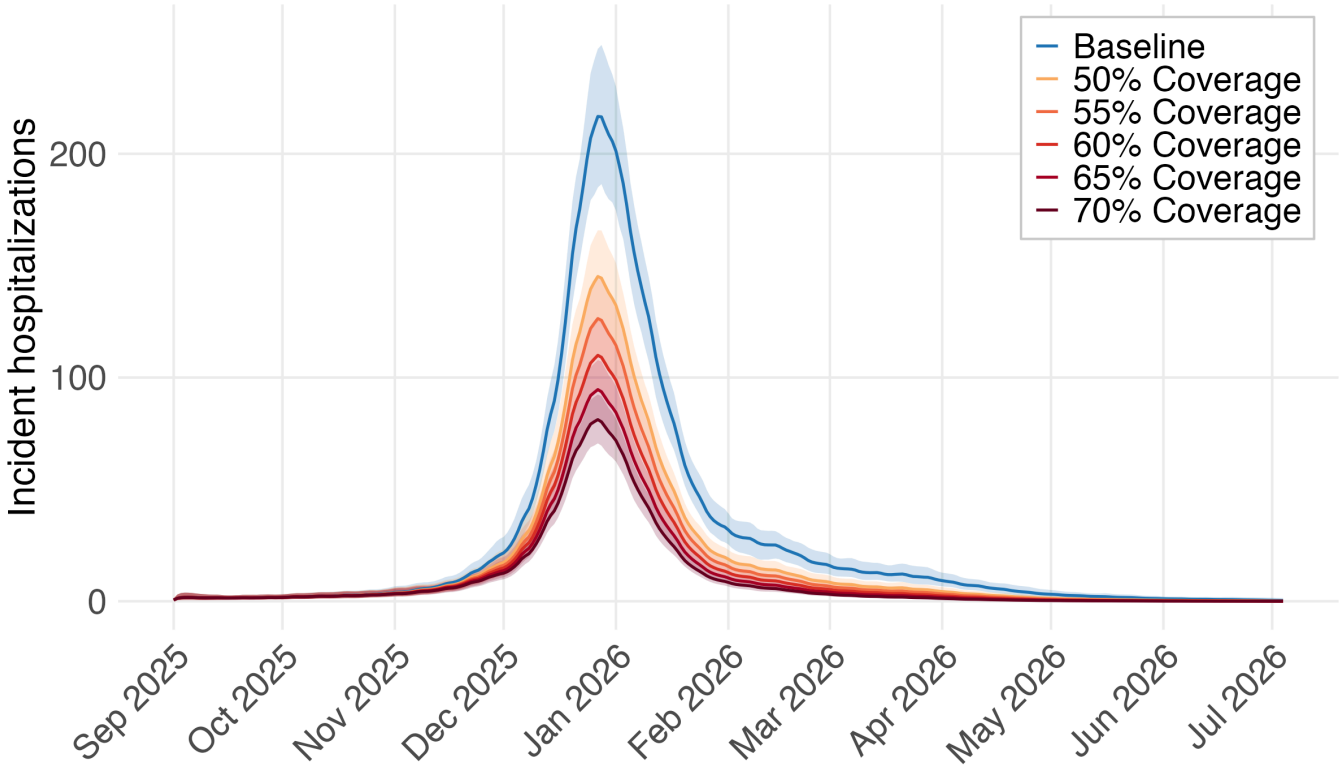
4.1 All age groups together

Scenario	Overall coverage	Doses vs. baseline	Hospitalizations	Change vs. baseline	% change
No vaccination	0.0%	-2,809,361	22,545 [19,359 – 25,445]	+14,346 [+11,362 – +16,901]	+173.1% [+138.2% – +200.2%]
-20%	30.9%	-561,872	10,312 [9,585 – 11,034]	+2,049 [+1,773 – +2,260]	+24.8% [+22.0% – +26.7%]
-15%	32.8%	-421,404	9,757 [9,089 – 10,431]	+1,501 [+1,308 – +1,647]	+18.2% [+16.2% – +19.4%]
-10%	34.8%	-280,936	9,237 [8,610 – 9,845]	+976 [+856 – +1,069]	+11.8% [+10.6% – +12.6%]
-5%	36.7%	-140,468	8,741 [8,152 – 9,309]	+476 [+419 – +519]	+5.8% [+5.2% – +6.1%]
Baseline	38.6%	+0	8,268 [7,715 – 8,802]	+0 [+0 – +0]	+0.0% [+0.0% – +0.0%]
50% target	52.1%	+981,541	5,413 [5,041 – 5,828]	-2,860 [-3,087 – -2,599]	-34.6% [-36.1% – -32.4%]
55% target	56.1%	+1,273,763	4,687 [4,355 – 5,085]	-3,589 [-3,865 – -3,283]	-43.4% [-45.0% – -41.0%]
60% target	60.2%	+1,566,373	4,059 [3,764 – 4,411]	-4,219 [-4,532 – -3,874]	-51.0% [-52.6% – -48.6%]
65% target	65.0%	+1,918,046	3,487 [3,224 – 3,801]	-4,787 [-5,134 – -4,416]	-57.9% [-59.5% – -55.6%]
70% target	70.0%	+2,281,693	2,995 [2,758 – 3,262]	-5,279 [-5,665 – -4,881]	-63.8% [-65.3% – -61.7%]

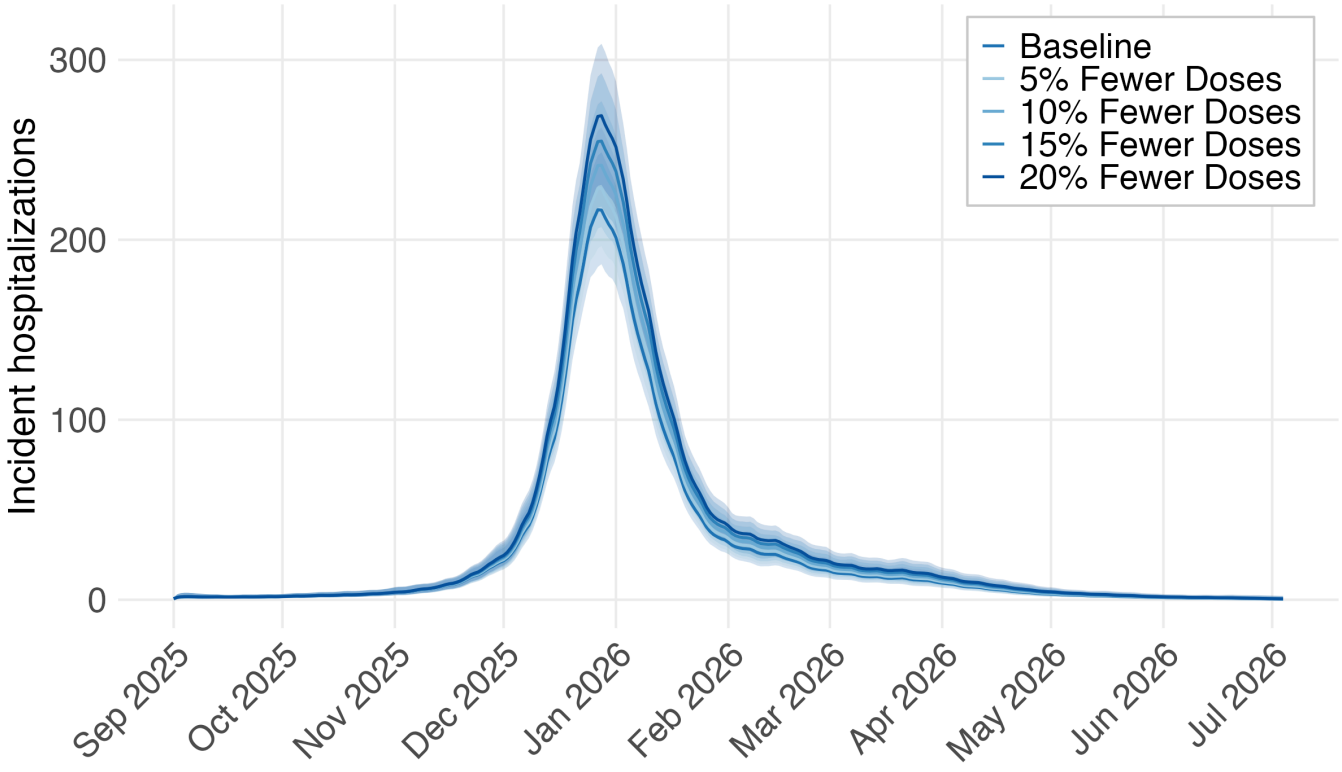
Overall coverage = scheduled doses summed over all age groups ÷ total population, i.e. the population-weighted average of each age group's scheduled coverage.



Season hospitalizations against overall scheduled coverage



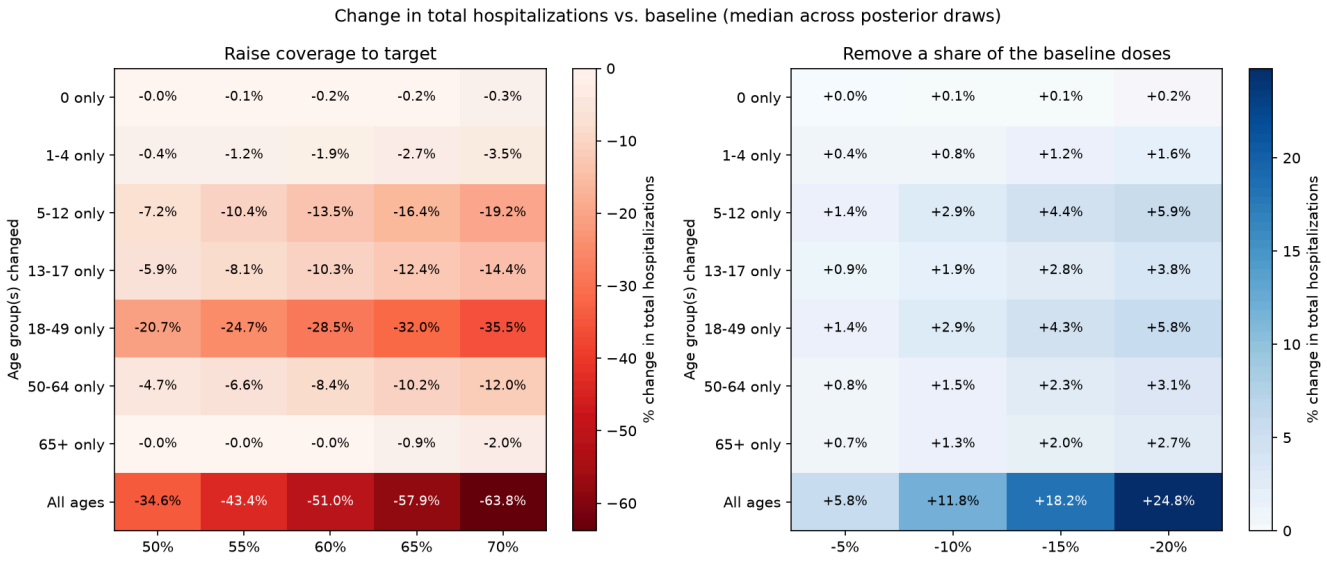
Daily hospitalizations: baseline and coverage targets (all ages)



Daily hospitalizations: baseline and fewer doses (all ages)

4.2 One age group at a time

Change in **total** (all-ages) hospitalizations when only the row's age group is changed, median across draws (95% intervals in appendix A):



% change in total hospitalizations, by age group changed and level

Hospitalizations averted vs. baseline, all ages counted

Raised to target	50%	55%	60%	65%	70%
0 only	0 [0 – 0]	6 [5 – 6]	13 [12 – 14]	19 [18 – 21]	26 [24 – 28]
1-4 only	30 [26 – 32]	95 [85 – 104]	161 [143 – 175]	225 [200 – 246]	289 [257 – 315]
5-12 only	593 [520 – 651]	859 [756 – 940]	1,112 [982 – 1,215]	1,354 [1,199 – 1,477]	1,587 [1,410 – 1,730]
13-17 only	488 [425 – 533]	672 [589 – 735]	851 [747 – 930]	1,025 [901 – 1,119]	1,193 [1,050 – 1,302]
18-49 only	1,704 [1,536 – 1,850]	2,034 [1,841 – 2,205]	2,349 [2,128 – 2,539]	2,647 [2,401 – 2,857]	2,927 [2,669 – 3,159]
50-64 only	385 [341 – 421]	541 [480 – 592]	693 [617 – 758]	844 [752 – 921]	990 [884 – 1,081]
65+ only	0 [0 – 0]	0 [0 – 0]	0 [0 – 0]	75 [68 – 82]	165 [149 – 180]
All ages	2,860 [2,599 – 3,087]	3,589 [3,283 – 3,865]	4,219 [3,874 – 4,532]	4,787 [4,416 – 5,134]	5,279 [4,881 – 5,665]

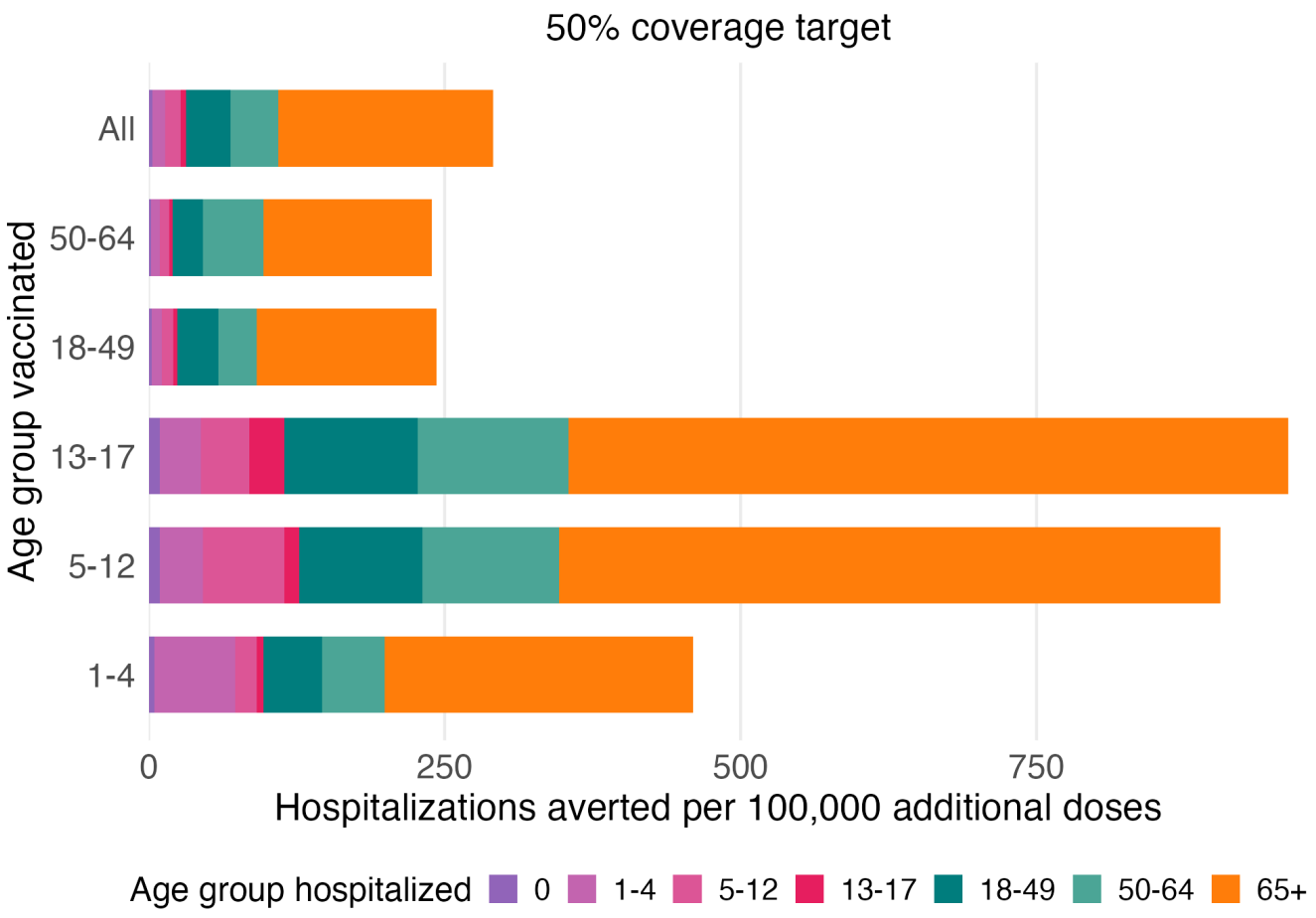
Additional hospitalizations vs. baseline, all ages counted

Doses cut	–5%	–10%	–15%	–20%
0 only	3 [3 – 4]	7 [6 – 7]	10 [9 – 11]	14 [12 – 15]
1-4 only	32 [28 – 35]	64 [57 – 70]	96 [85 – 104]	128 [114 – 140]
5-12 only	118 [102 – 130]	239 [206 – 263]	361 [312 – 399]	486 [418 – 537]
13-17 only	77 [67 – 85]	156 [134 – 171]	235 [202 – 259]	316 [271 – 347]
18-49 only	117 [103 – 129]	236 [206 – 259]	356 [311 – 392]	479 [416 – 526]
50-64 only	64 [56 – 70]	128 [112 – 140]	192 [169 – 211]	257 [226 – 283]
65+ only	55 [50 – 60]	110 [99 – 119]	165 [149 – 179]	220 [198 – 239]
All ages	476 [419 – 519]	976 [856 – 1,069]	1,501 [1,308 – 1,647]	2,049 [1,773 – 2,260]

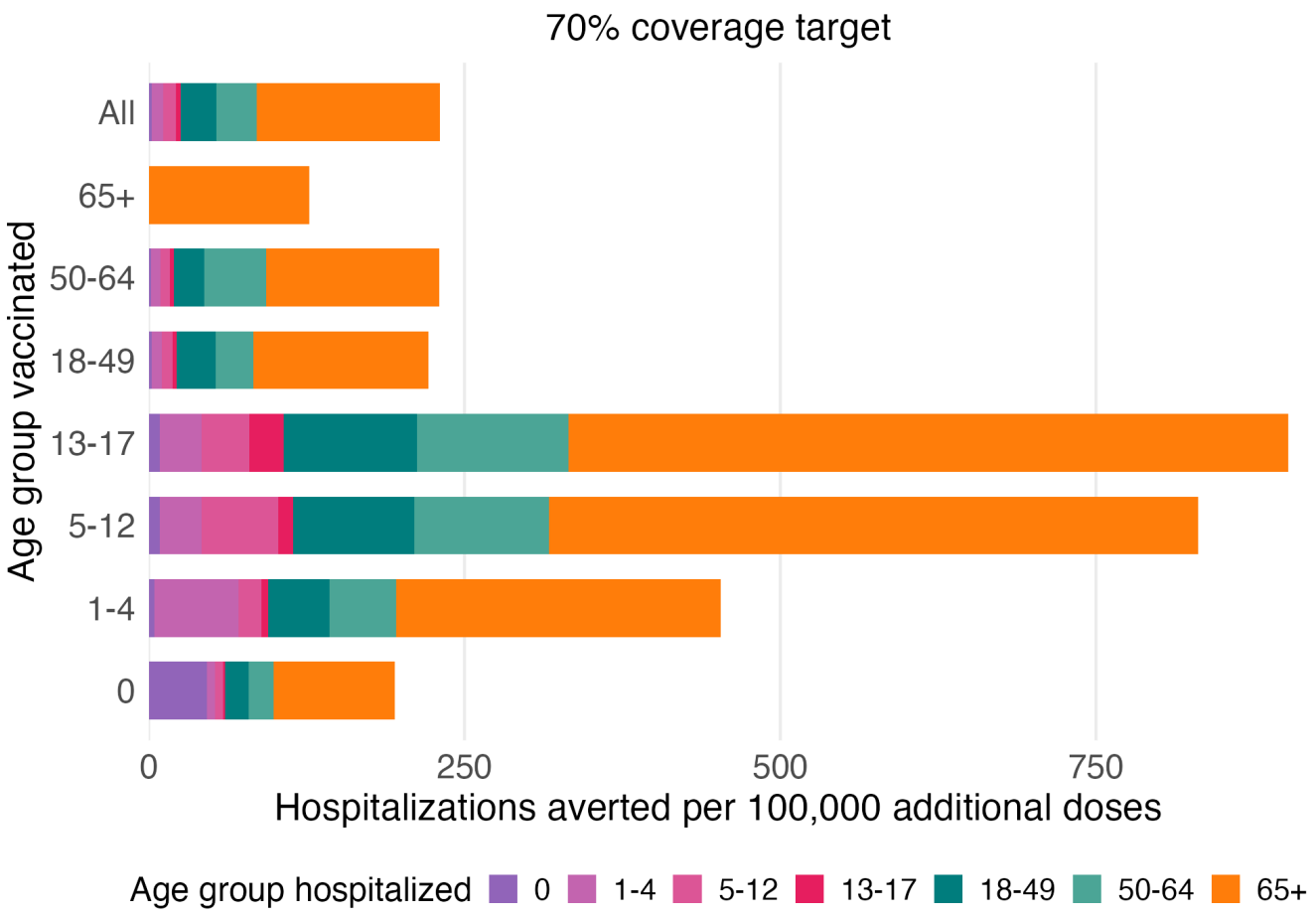
4.3 Efficiency per dose

Hospitalizations averted per 100,000 **additional** doses (targets) and added per 100,000 doses **removed** (cuts), all ages counted. For targets the denominator is the extra doses the target requires in that group (`All ages`: in every group).

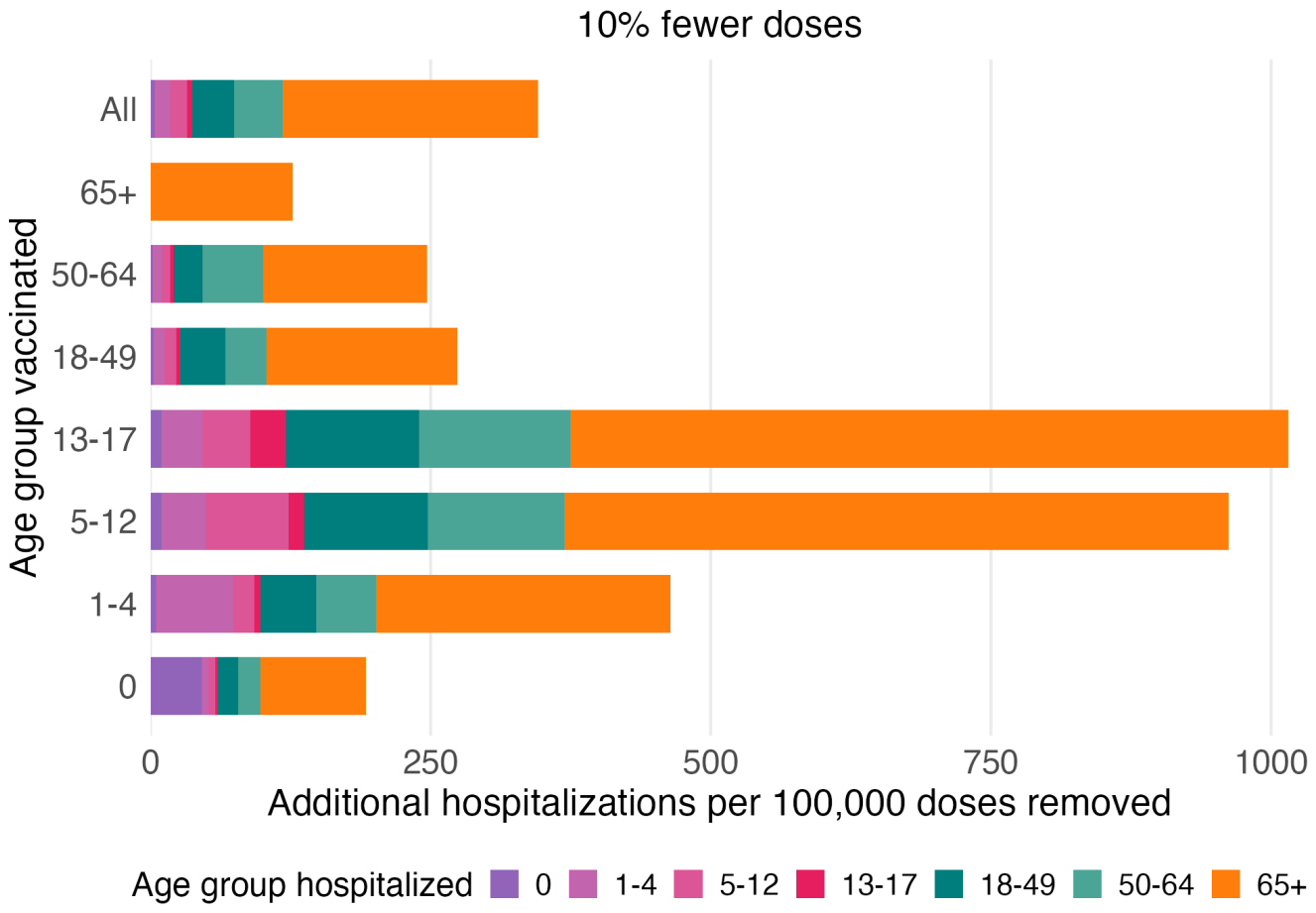
Age group changed	50% target	55% target	60% target	65% target	70% target	–5%	–10%	–15%	–20%
0 only	—	196.7 [179.4 – 211.1]	195.9 [178.7 – 210.0]	196.0 [178.9 – 210.3]	195.7 [178.5 – 209.9]	194.9 [177.7 – 209.2]	193.3 [176.6 – 207.4]	195.8 [178.9 – 210.1]	195.1 [178.0 – 209.2]
1-4 only	460.4 [409.0 – 501.7]	458.9 [408.0 – 500.8]	457.2 [406.5 – 499.0]	454.8 [404.5 – 496.3]	452.6 [402.5 – 493.8]	462.5 [411.3 – 505.1]	464.1 [413.5 – 507.1]	465.1 [414.4 – 508.2]	466.8 [415.4 – 509.7]
5-12 only	907.0 [794.6 – 994.5]	887.8 [782.1 – 971.5]	868.9 [767.7 – 949.4]	850.5 [753.0 – 927.6]	832.8 [739.8 – 907.9]	957.2 [828.8 – 1054.6]	964.7 [833.9 – 1064.3]	973.7 [840.2 – 1075.1]	981.7 [845.6 – 1085.1]
13-17 only	967.1 [843.0 – 1056.4]	950.2 [831.8 – 1038.0]	934.3 [820.0 – 1020.7]	919.1 [808.5 – 1003.9]	905.2 [796.6 – 987.7]	1011.1 [870.8 – 1108.1]	1020.4 [877.2 – 1119.3]	1025.7 [881.1 – 1126.6]	1030.9 [885.3 – 1133.8]
18-49 only	243.8 [219.8 – 264.7]	238.1 [215.5 – 258.1]	232.5 [210.7 – 251.4]	227.0 [206.0 – 245.1]	221.5 [202.0 – 239.0]	273.6 [239.2 – 300.3]	275.3 [240.4 – 302.4]	277.1 [241.5 – 304.6]	279.1 [242.7 – 306.9]
50-64 only	239.7 [212.4 – 262.3]	237.1 [210.7 – 259.5]	234.6 [208.8 – 256.5]	232.3 [207.0 – 253.7]	229.9 [205.2 – 250.9]	246.8 [217.7 – 270.8]	247.7 [218.3 – 271.8]	248.6 [218.9 – 273.0]	249.6 [219.6 – 274.2]
65+ only	—	—	—	127.0 [114.7 – 138.0]	127.0 [114.7 – 138.0]	127.0 [114.7 – 138.0]	127.0 [114.7 – 138.0]	127.0 [114.7 – 138.0]	127.0 [114.7 – 138.0]
All ages	291.4 [264.8 – 314.6]	281.8 [257.8 – 303.5]	269.3 [247.3 – 289.3]	249.6 [230.2 – 267.7]	231.4 [213.9 – 248.3]	338.8 [298.2 – 369.4]	347.6 [304.6 – 380.4]	356.1 [310.3 – 390.9]	364.6 [315.5 – 402.2]



50% target: hospitalizations averted per 100,000 additional doses, by age group hospitalized



70% target: hospitalizations averted per 100,000 additional doses, by age group hospitalized



10% fewer doses: additional hospitalizations per 100,000 doses removed

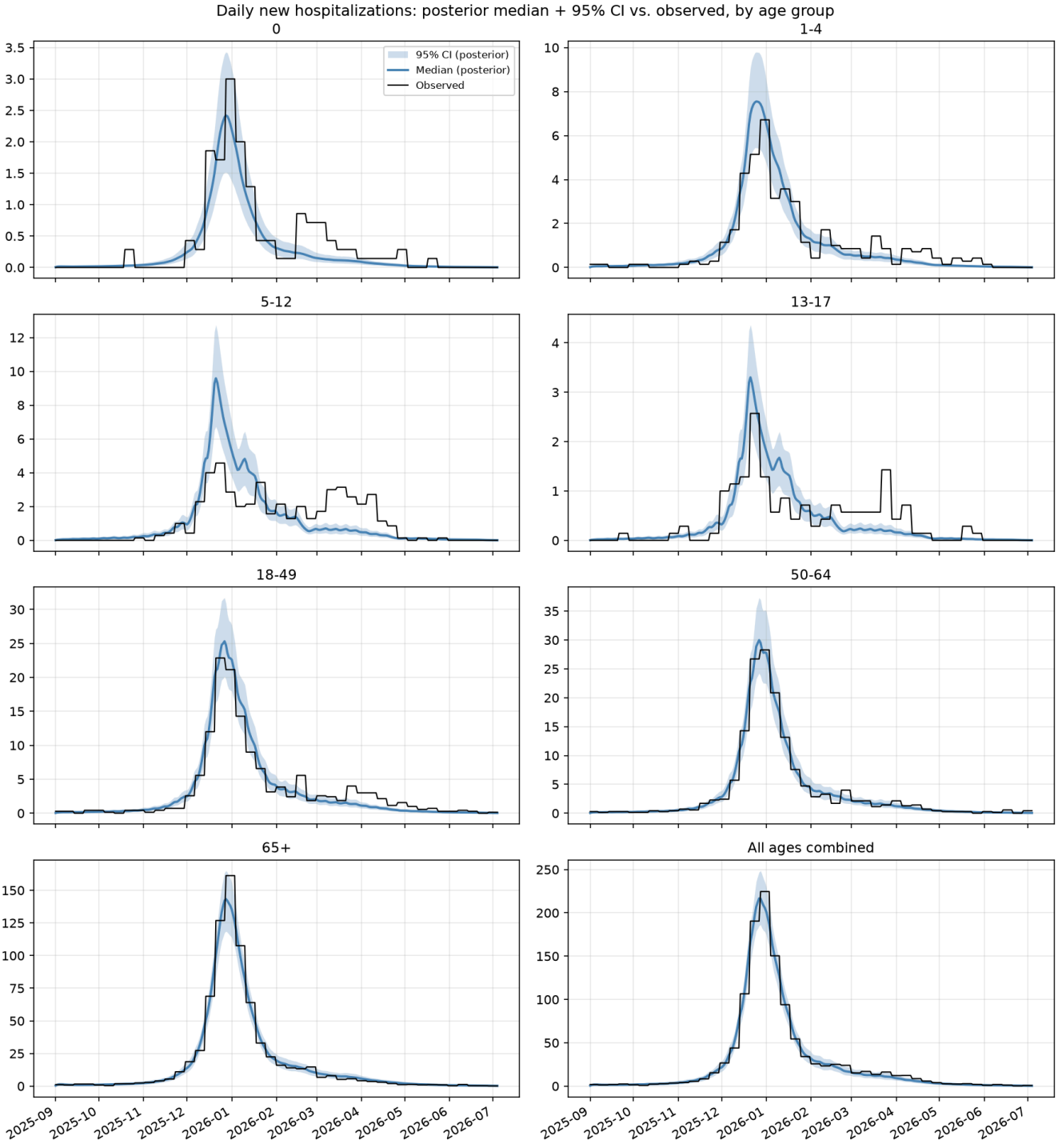
A relative dose cut removes the most doses where coverage is highest, so per removed dose its cost mirrors the per-dose benefit of Table S.A.2; coverage targets add doses mostly to the least-covered groups (18-49 needs the largest relative increase).

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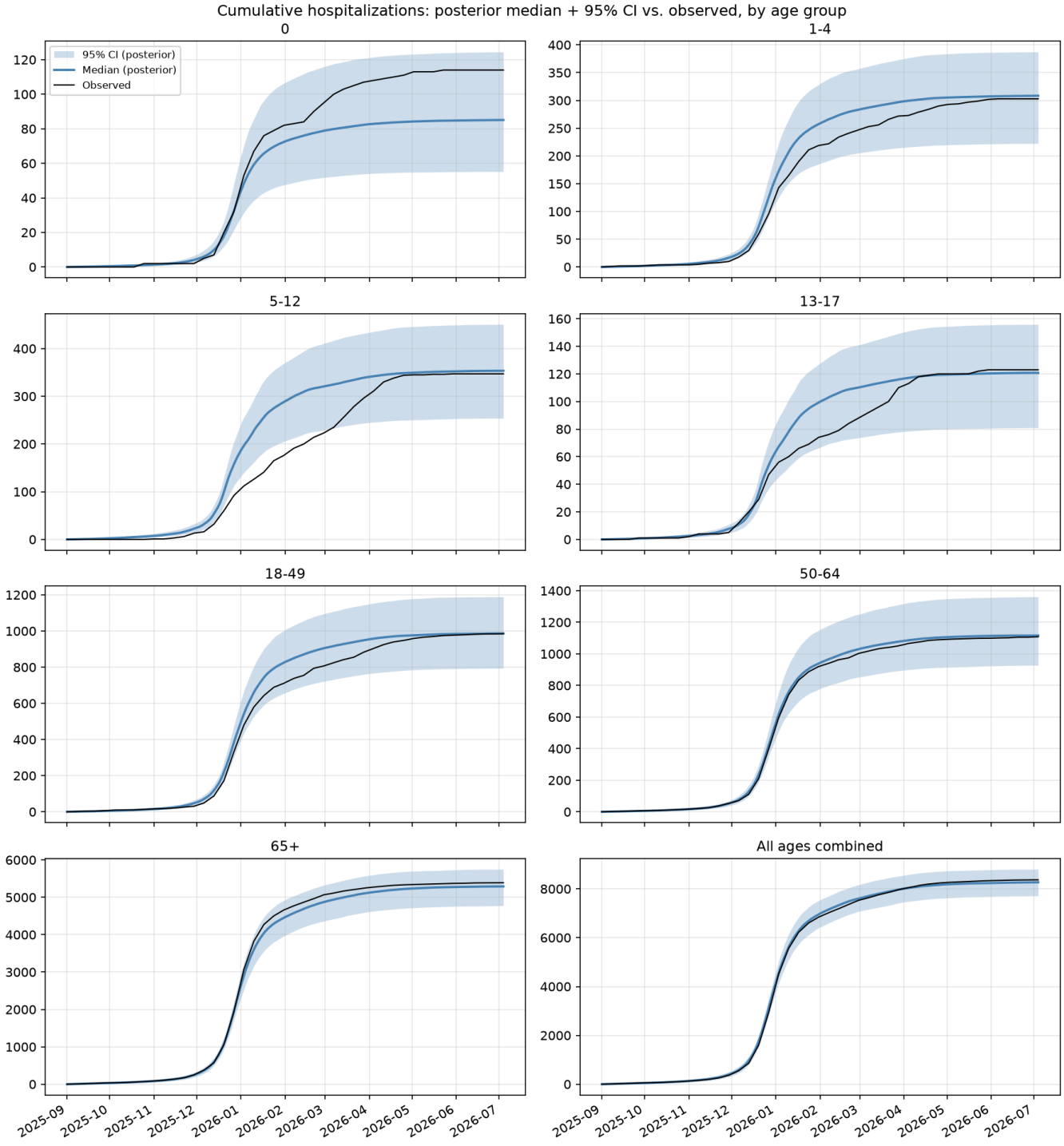
5. Baseline fit check

Cumulative admissions over 2025-09-01 – 2026-07-04:

Age group	Simulated (median)	Simulated 95% interval	Observed	% difference
0	85.1	55.2 – 124.4	114.0	-25.4%
1-4	308.3	222.4 – 387.1	302.9	+1.8%
5-12	353.5	253.9 – 450.6	347.0	+1.9%
13-17	120.8	81.0 – 155.9	123.0	-1.8%
18-49	986.7	794.1 – 1,188.9	984.7	+0.2%
50-64	1,116.1	927.5 – 1,361.1	1,108.7	+0.7%
65+	5,290.8	4,776.1 – 5,750.6	5,388.9	-1.8%
Total	**8,268.0**	**7,714.8 – 8,802.1**	**8,369.1**	**-1.2%**



Daily admissions: posterior median and 95% band vs. observed



Cumulative admissions: posterior median and 95% band vs. observed

6. Data sources

Input	Source
Hospital admissions (calibration target)	Massachusetts Department of Public Health (MA DPH): weekly influenza admissions by county and age group (`influenza_timeseries_*.csv`, `ed_admits_count`), summed over counties and spread evenly over each MMWR week (`data/prepare_hospitalizations.py`)
Vaccination	MA DPH: weekly cumulative first doses by county and age (`MA_influenza_coverage_2025-2026.csv`), differenced and spread over each week (`data/prepare_vaccination.py`)
Population by age group	MA DPH: `POP_ESTIMATE` of the vaccination file, summed over counties
Contact matrices	Mistry et al. 2021 synthetic contact matrices
Absolute humidity	gridMET daily specific humidity averaged over Massachusetts (× 1.2 kg/m³, `data/extract_ma_humidity.py`)
School/work calendar	Constructed state school/work-day calendar

7. Notes

- Intervals reflect calibration uncertainty (posterior draws, deterministic transitions), not epidemic-process noise.
- `m(t)` is a statistical smoothing device, not an independently measured behavioural signal.
- Coverage-target multipliers are computed from the scheduled coverage over the simulated window; the model's cap means delivered coverage lands slightly below the target (appendix B).
- "Per 100,000 doses" uses scheduled doses: doses the model declines to deliver to already-infected people are still bought and given.
- Vaccinating 65+ has no indirect effect on other age groups because vaccination blocks no infection in 65+ in the fitted model (residual susceptibility 1.00).

Appendix A — Coverage-scenario tables by age group hospitalized

50% coverage target — hospitalizations averted vs. baseline

Counted in	0 raised	1-4 raised	5-12 raised	13-17 raised	18-49 raised	50-64 raised	65+ raised	All raised
0	0 [0 – 0]	0 [0 – 0]	6 [4 – 8]	5 [3 – 7]	18 [12 – 24]	3 [2 – 4]	0 [0 – 0]	29 [19 – 40]
1-4	0 [0 – 0]	4 [3 – 5]	24 [17 – 30]	18 [12 – 22]	62 [45 – 78]	12 [9 – 15]	0 [0 – 0]	106 [77 – 133]
5-12	0 [0 – 0]	1 [1 – 2]	45 [32 – 58]	20 [15 – 27]	66 [47 – 87]	13 [9 – 17]	0 [0 – 0]	128 [92 – 166]
13-17	0 [0 – 0]	0 [0 – 0]	8 [6 – 11]	15 [10 – 19]	23 [15 – 30]	5 [3 – 6]	0 [0 – 0]	45 [30 – 59]
18-49	0 [0 – 0]	3 [3 – 4]	68 [53 – 84]	57 [45 – 70]	243 [195 – 296]	40 [32 – 49]	0 [0 – 0]	368 [294 – 446]
50-64	0 [0 – 0]	3 [3 – 4]	75 [60 – 94]	64 [51 – 81]	226 [183 – 281]	83 [68 – 102]	0 [0 – 0]	401 [327 – 494]
65+	0 [0 – 0]	17 [15 – 19]	366 [316 – 408]	307 [265 – 342]	1,063 [944 – 1,176]	228 [200 – 253]	0 [0 – 0]	1,779 [1,592 – 1,959]
All	**0 [0 – 0]**	**30 [26 – 32]**	**593 [520 – 651]**	**488 [425 – 533]**	**1,704 [1,536 – 1,850]**	**385 [341 – 421]**	**0 [0 – 0]**	**2,860 [2,599 – 3,087]**

55% coverage target — hospitalizations averted vs. baseline

Counted in	0 raised	1-4 raised	5-12 raised	13-17 raised	18-49 raised	50-64 raised	65+ raised	All raised
0	1 [1 – 2]	1 [1 – 1]	9 [6 – 12]	7 [5 – 9]	21 [14 – 29]	5 [3 – 6]	0 [0 – 0]	37 [25 – 52]
1-4	0 [0 – 0]	14 [10 – 18]	34 [25 – 43]	24 [17 – 30]	74 [53 – 93]	17 [12 – 21]	0 [0 – 0]	137 [99 – 172]
5-12	0 [0 – 0]	4 [3 – 5]	64 [46 – 84]	28 [20 – 37]	79 [57 – 103]	18 [13 – 24]	0 [0 – 0]	162 [117 – 209]
13-17	0 [0 – 0]	1 [1 – 2]	12 [8 – 16]	20 [14 – 27]	27 [18 – 36]	6 [4 – 9]	0 [0 – 0]	56 [37 – 73]
18-49	1 [0 – 1]	10 [8 – 13]	98 [78 – 122]	78 [62 – 97]	289 [232 – 351]	57 [45 – 69]	0 [0 – 0]	456 [366 – 550]
50-64	1 [0 – 1]	11 [9 – 14]	109 [87 – 136]	89 [71 – 111]	271 [219 – 335]	116 [95 – 143]	0 [0 – 0]	502 [412 – 616]
65+	3 [3 – 3]	54 [47 – 60]	529 [459 – 589]	423 [366 – 471]	1,270 [1,133 – 1,401]	321 [283 – 356]	0 [0 – 0]	2,233 [2,005 – 2,455]
All	**6 [5 – 6]**	**95 [85 – 104]**	**859 [756 – 940]**	**672 [589 – 735]**	**2,034 [1,841 – 2,205]**	**541 [480 – 592]**	**0 [0 – 0]**	**3,589 [3,283 – 3,865]**

60% coverage target — hospitalizations averted vs. baseline

Counted in	0 raised	1-4 raised	5-12 raised	13-17 raised	18-49 raised	50-64 raised	65+ raised	All raised
0	3 [2 – 4]	2 [1 – 2]	11 [8 – 16]	8 [6 – 11]	24 [16 – 33]	6 [4 – 8]	0 [0 – 0]	44 [29 – 62]
1-4	0 [0 – 1]	24 [17 – 30]	44 [32 – 56]	31 [22 – 39]	85 [62 – 107]	21 [15 – 27]	0 [0 – 0]	163 [118 – 204]
5-12	0 [0 – 1]	6 [5 – 9]	83 [60 – 107]	36 [26 – 47]	92 [66 – 120]	23 [17 – 31]	0 [0 – 0]	191 [137 – 246]
13-17	0 [0 – 0]	2 [1 – 3]	16 [11 – 21]	26 [17 – 33]	31 [21 – 42]	8 [6 – 11]	0 [0 – 0]	66 [44 – 85]
18-49	1 [1 – 1]	17 [14 – 21]	128 [101 – 157]	99 [79 – 123]	333 [267 – 403]	73 [58 – 89]	0 [0 – 0]	531 [427 – 640]
50-64	1 [1 – 2]	19 [15 – 23]	142 [114 – 177]	113 [90 – 141]	313 [254 – 386]	148 [122 – 183]	0 [0 – 0]	590 [483 – 720]
65+	6 [5 – 7]	91 [79 – 101]	686 [598 – 763]	537 [466 – 597]	1,466 [1,311 – 1,617]	412 [364 – 456]	0 [0 – 0]	2,622 [2,363 – 2,880]
All	**13 [12 – 14]**	**161 [143 – 175]**	**1,112 [982 – 1,215]**	**851 [747 – 930]**	**2,349 [2,128 – 2,539]**	**693 [617 – 758]**	**0 [0 – 0]**	**4,219 [3,874 – 4,532]**

65% coverage target — hospitalizations averted vs. baseline

Counted in	0 raised	1-4 raised	5-12 raised	13-17 raised	18-49 raised	50-64 raised	65+ raised	All raised
0	5 [3 – 7]	2 [2 – 3]	14 [9 – 19]	10 [7 – 14]	27 [18 – 38]	7 [5 – 10]	0 [-0 – 0]	50 [33 – 71]
1-4	1 [0 – 1]	33 [24 – 41]	54 [39 – 68]	37 [27 – 46]	96 [70 – 120]	26 [19 – 33]	0 [-0 – 0]	185 [134 – 232]
5-12	1 [0 – 1]	9 [6 – 12]	100 [72 – 129]	43 [31 – 57]	103 [74 – 135]	28 [20 – 38]	0 [-0 – 0]	215 [155 – 277]
13-17	0 [0 – 0]	3 [2 – 4]	19 [13 – 25]	31 [20 – 40]	36 [24 – 47]	10 [7 – 13]	0 [-0 – 0]	74 [49 – 96]
18-49	2 [1 – 2]	24 [19 – 30]	155 [123 – 191]	120 [95 – 148]	374 [300 – 451]	89 [71 – 108]	0 [-0 – 0]	594 [479 – 716]
50-64	2 [2 – 2]	26 [21 – 33]	173 [139 – 215]	136 [109 – 169]	353 [287 – 435]	180 [148 – 221]	0 [-0 – 0]	663 [545 – 810]
65+	10 [8 – 11]	128 [111 – 142]	836 [730 – 928]	646 [562 – 718]	1,654 [1,478 – 1,819]	502 [443 – 555]	75 [68 – 82]	2,995 [2,711 – 3,286]
All	**19 [18 – 21]**	**225 [200 – 246]**	**1,354 [1,199 – 1,477]**	**1,025 [901 – 1,119]**	**2,647 [2,401 – 2,857]**	**844 [752 – 921]**	**75 [68 – 82]**	**4,787 [4,416 – 5,134]**

70% coverage target — hospitalizations averted vs. baseline

Counted in	0 raised	1-4 raised	5-12 raised	13-17 raised	18-49 raised	50-64 raised	65+ raised	All raised
0	6 [4 – 9]	3 [2 – 4]	16 [11 – 22]	12 [8 – 16]	30 [20 – 42]	9 [6 – 12]	0 [-0 – 0]	55 [36 – 78]
1-4	1 [1 – 1]	42 [31 – 53]	63 [46 – 79]	43 [31 – 54]	106 [77 – 133]	31 [22 – 39]	0 [-0 – 0]	204 [147 – 255]
5-12	1 [1 – 1]	12 [8 – 15]	116 [84 – 150]	50 [36 – 66]	115 [82 – 149]	33 [24 – 44]	0 [-0 – 0]	236 [169 – 303]
13-17	0 [0 – 0]	3 [2 – 5]	23 [15 – 30]	35 [23 – 46]	39 [26 – 52]	12 [8 – 16]	0 [-0 – 0]	80 [54 – 105]
18-49	2 [2 – 3]	31 [25 – 38]	182 [144 – 224]	139 [110 – 172]	412 [331 – 498]	105 [83 – 127]	0 [-0 – 0]	648 [523 – 783]
50-64	3 [2 – 3]	34 [27 – 42]	203 [162 – 252]	158 [127 – 197]	391 [318 – 483]	210 [173 – 259]	0 [-0 – 0]	727 [598 – 886]
65+	13 [11 – 14]	164 [143 – 182]	980 [858 – 1,087]	751 [655 – 835]	1,830 [1,642 – 2,011]	589 [522 – 652]	165 [149 – 180]	3,318 [3,003 – 3,634]
All	**26 [24 – 28]**	**289 [257 – 315]**	**1,587 [1,410 – 1,730]**	**1,193 [1,050 – 1,302]**	**2,927 [2,669 – 3,159]**	**990 [884 – 1,081]**	**165 [149 – 180]**	**5,279 [4,881 – 5,665]**

5% fewer doses — additional hospitalizations vs. baseline

Counted in	0 cut	1-4 cut	5-12 cut	13-17 cut	18-49 cut	50-64 cut	65+ cut	All cut
0	1 [1 – 1]	0 [0 – 0]	1 [1 – 2]	1 [1 – 1]	1 [1 – 2]	1 [0 – 1]	0 [-0 – 0]	5 [3 – 7]
1-4	0 [0 – 0]	5 [3 – 6]	5 [3 – 6]	3 [2 – 3]	4 [3 – 5]	2 [1 – 2]	0 [-0 – 0]	19 [14 – 24]
5-12	0 [0 – 0]	1 [1 – 2]	9 [7 – 12]	3 [2 – 4]	4 [3 – 6]	2 [2 – 3]	0 [-0 – 0]	21 [15 – 27]
13-17	0 [0 – 0]	0 [0 – 1]	2 [1 – 2]	2 [2 – 3]	2 [1 – 2]	1 [1 – 1]	0 [-0 – 0]	7 [5 – 9]
18-49	0 [0 – 0]	3 [3 – 4]	14 [11 – 17]	9 [7 – 11]	17 [14 – 21]	7 [5 – 8]	0 [-0 – 0]	51 [40 – 63]
50-64	0 [0 – 0]	4 [3 – 5]	15 [12 – 19]	10 [8 – 13]	15 [12 – 19]	14 [11 – 17]	0 [-0 – 0]	59 [47 – 74]
65+	2 [1 – 2]	18 [16 – 20]	73 [62 – 81]	49 [42 – 54]	73 [63 – 81]	38 [33 – 42]	55 [50 – 60]	312 [274 – 347]
All	**3 [3 – 4]**	**32 [28 – 35]**	**118 [102 – 130]**	**77 [67 – 85]**	**117 [103 – 129]**	**64 [56 – 70]**	**55 [50 – 60]**	**476 [419 – 519]**

10% fewer doses — additional hospitalizations vs. baseline

Counted in	0 cut	1-4 cut	5-12 cut	13-17 cut	18-49 cut	50-64 cut	65+ cut	All cut
0	2 [1 – 2]	1 [0 – 1]	2 [2 – 3]	2 [1 – 2]	2 [2 – 3]	1 [1 – 1]	0 [-0 – 0]	10 [7 – 14]
1-4	0 [0 – 0]	9 [7 – 12]	10 [7 – 12]	6 [4 – 7]	9 [6 – 11]	4 [3 – 5]	0 [-0 – 0]	39 [28 – 49]
5-12	0 [0 – 0]	3 [2 – 3]	18 [13 – 24]	7 [5 – 9]	9 [6 – 12]	4 [3 – 6]	0 [-0 – 0]	42 [30 – 56]
13-17	0 [0 – 0]	1 [1 – 1]	3 [2 – 4]	5 [3 – 6]	3 [2 – 4]	2 [1 – 2]	0 [-0 – 0]	14 [9 – 19]
18-49	1 [1 – 1]	7 [5 – 8]	27 [21 – 34]	18 [14 – 22]	34 [27 – 42]	13 [10 – 16]	0 [-0 – 0]	104 [82 – 128]
50-64	1 [1 – 1]	7 [6 – 9]	30 [24 – 38]	21 [16 – 26]	31 [25 – 39]	28 [23 – 34]	0 [-0 – 0]	121 [97 – 152]
65+	3 [3 – 4]	36 [31 – 40]	147 [125 – 164]	98 [84 – 109]	146 [127 – 163]	75 [66 – 84]	110 [99 – 119]	641 [559 – 713]
All	**7 [6 – 7]**	**64 [57 – 70]**	**239 [206 – 263]**	**156 [134 – 171]**	**236 [206 – 259]**	**128 [112 – 140]**	**110 [99 – 119]**	**976 [856 – 1,069]**

15% fewer doses — additional hospitalizations vs. baseline

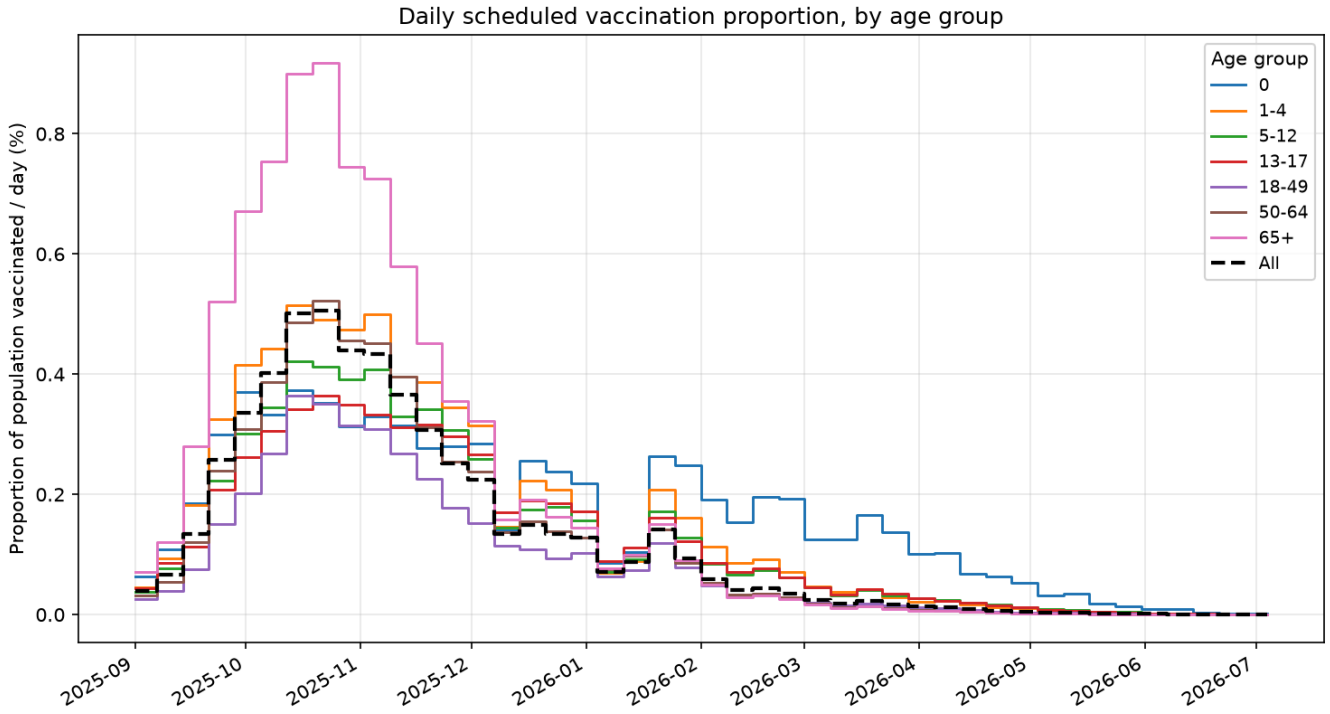
Counted in	0 cut	1-4 cut	5-12 cut	13-17 cut	18-49 cut	50-64 cut	65+ cut	All cut
0	2 [2 – 3]	1 [1 – 1]	4 [3 – 5]	2 [2 – 3]	4 [3 – 5]	2 [1 – 2]	0 [-0 – 0]	16 [11 – 21]
1-4	0 [0 – 0]	14 [10 – 18]	15 [10 – 18]	8 [6 – 11]	13 [9 – 16]	6 [4 – 7]	0 [-0 – 0]	60 [43 – 75]
5-12	0 [0 – 0]	4 [3 – 5]	28 [20 – 37]	10 [7 – 13]	14 [10 – 18]	6 [5 – 8]	0 [-0 – 0]	65 [47 – 86]
13-17	0 [0 – 0]	1 [1 – 2]	5 [3 – 7]	7 [5 – 10]	5 [3 – 6]	2 [2 – 3]	0 [-0 – 0]	22 [14 – 29]
18-49	1 [1 – 1]	10 [8 – 13]	41 [32 – 51]	27 [21 – 34]	52 [41 – 64]	20 [16 – 25]	0 [-0 – 0]	159 [125 – 197]
50-64	1 [1 – 1]	11 [9 – 14]	46 [36 – 57]	31 [24 – 39]	47 [37 – 59]	42 [34 – 52]	0 [-0 – 0]	186 [147 – 234]
65+	5 [4 – 6]	54 [47 – 60]	222 [189 – 248]	148 [126 – 165]	221 [191 – 246]	114 [99 – 126]	165 [149 – 179]	987 [854 – 1,097]
All	**10 [9 – 11]**	**96 [85 – 104]**	**361 [312 – 399]**	**235 [202 – 259]**	**356 [311 – 392]**	**192 [169 – 211]**	**165 [149 – 179]**	**1,501 [1,308 – 1,647]**

20% fewer doses — additional hospitalizations vs. baseline

Counted in	0 cut	1-4 cut	5-12 cut	13-17 cut	18-49 cut	50-64 cut	65+ cut	All cut
0	3 [2 – 5]	1 [1 – 2]	5 [3 – 7]	3 [2 – 4]	5 [3 – 7]	2 [2 – 3]	0 [-0 – 0]	21 [15 – 29]
1-4	0 [0 – 1]	19 [14 – 24]	20 [14 – 25]	11 [8 – 14]	17 [12 – 22]	8 [6 – 10]	0 [-0 – 0]	82 [58 – 103]
5-12	0 [0 – 1]	5 [4 – 7]	38 [27 – 49]	13 [9 – 18]	18 [13 – 24]	9 [6 – 11]	0 [-0 – 0]	89 [64 – 118]
13-17	0 [0 – 0]	2 [1 – 2]	7 [4 – 9]	10 [7 – 13]	6 [4 – 8]	3 [2 – 4]	0 [-0 – 0]	30 [20 – 39]
18-49	1 [1 – 2]	14 [11 – 17]	55 [43 – 69]	37 [29 – 45]	70 [55 – 86]	27 [21 – 33]	0 [-0 – 0]	216 [170 – 269]
50-64	1 [1 – 2]	15 [12 – 18]	61 [48 – 77]	42 [33 – 52]	63 [50 – 79]	56 [46 – 69]	0 [-0 – 0]	254 [199 – 320]
65+	7 [6 – 7]	72 [63 – 80]	298 [254 – 334]	198 [169 – 222]	297 [256 – 330]	152 [133 – 169]	220 [198 – 239]	1,350 [1,162 – 1,503]
All	**14 [12 – 15]**	**128 [114 – 140]**	**486 [418 – 537]**	**316 [271 – 347]**	**479 [416 – 526]**	**257 [226 – 283]**	**220 [198 – 239]**	**2,049 [1,773 – 2,260]**

Appendix B — Dose accounting: scheduled vs. delivered doses

Age group	Population	Scheduled doses	Delivered doses	Wasted doses	Pct wasted	Scheduled coverage	Delivered coverage
0	68,723	34,751	34,204	547	1.6%	50.6%	49.8%
1-4	287,051	137,081	135,736	1,345	1.0%	47.8%	47.3%
5-12	625,439	247,286	243,217	4,069	1.6%	39.5%	38.9%
13-17	406,983	153,066	150,201	2,865	1.9%	37.6%	36.9%
18-49	3,112,693	857,553	846,710	10,843	1.3%	27.6%	27.2%
50-64	1,351,323	515,217	510,708	4,510	0.9%	38.1%	37.8%
65+	1,420,722	864,407	861,194	3,213	0.4%	60.8%	60.6%
All	**7,272,934**	**2,809,361**	**2,781,970**	**27,391**	**1.0%**	**38.6%**	**38.3%**



Daily scheduled vaccination proportion

Appendix C — Attack rate by vaccination status (baseline)

Share of each group infected over the season, median [95% interval]. Vaccinated = vaccinated at any point in the season (delivered doses), with their infections after vaccination; unvaccinated = everyone else, seeded infections included. The model never vaccinates someone already infected, so an infection before a person's vaccination date counts as unvaccinated. This timing effect lowers the vaccinated attack rate even where vaccination blocks no infection (65+).

Age group	Share vaccinated	Vaccinated	Unvaccinated	Total
0	50.0%	1.8% [1.2–3.0%]	6.6% [4.6–10.4%]	4.2% [2.9–6.7%]
1-4	47.4%	2.7% [1.9–4.4%]	6.8% [4.8–10.9%]	4.9% [3.4–7.9%]
5-12	39.0%	4.0% [2.8–6.4%]	9.7% [6.9–15.3%]	7.5% [5.3–11.9%]
13-17	37.1%	4.0% [2.8–6.6%]	10.1% [7.2–15.9%]	7.9% [5.6–12.5%]
18-49	27.4%	4.9% [3.5–8.0%]	7.9% [5.6–12.6%]	7.1% [5.0–11.4%]
50-64	38.1%	4.4% [3.1–7.2%]	7.0% [4.9–11.2%]	6.0% [4.2–9.7%]
65+	61.3%	3.4% [2.4–5.6%]	4.4% [3.1–7.0%]	3.8% [2.7–6.2%]
All	**38.5%**	**4.1% [2.9–6.7%]**	**7.6% [5.3–12.0%]**	**6.2% [4.4–10.0%]**